US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250287613
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	KIM; Eunji

MEMORY DEVICE

Abstract

A memory device may include a first cell array chip including a first memory cell array, a first peripheral circuit chip including a first page buffer circuit electrically connected to the first memory cell array, an input/output (I/O) chip including a first pad circuit, a second pad circuit, a first I/O circuit electrically connected to the first pad circuit, and a second I/O circuit electrically connected to the second pad circuit. The memory device may further include a second cell array chip including a second memory cell array; and a second peripheral circuit chip including a second page buffer circuit electrically connected to the second memory cell array.

Inventors: **KIM; Eunji (Suwon-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

Family ID: **1000008381013**

Assignee: **SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**

Appl. No.: **19/016828**

Filed: **January 10, 2025**

Foreign Application Priority Data

KR	10-2024-0032839	Mar. 07, 2024

Publication Classification

Int. Cl.: **H10B80/00** (20230101); **H01L23/00** (20060101); **H01L25/065** (20230101); **H01L25/18** (20230101)

U.S. Cl.:

CPC **H10B80/00** (20230201); **H01L24/08** (20130101); **H01L25/0657** (20130101); **H01L25/18** (20130101); H01L2224/08145 (20130101); H01L2225/06541 (20130101);

H01L2924/1431 (20130101); H01L2924/14511 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0032839, filed on Mar. 7, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] Embodiments of the present disclosure relate to a memory device, and more particularly, to a memory device having a structure in which a plurality of chips are stacked in a vertical direction.

1. Brief Description of Related Art

[0003] Recently, with the growing tendency toward small-sized and highly efficient devices in the semiconductor industry, downscaling semiconductor chips is emerging as an important task. Against this background, there is an increasing demand for structural and functional improvements in memory semiconductor devices.

[0004] A vertical memory structure has become one of the important solutions to meet the demand. That is, in the vertical memory structure, as the number of stages in a memory cell array increases, an area occupied by the memory cell array, from among components included in a memory device, may comparatively reduce. However, due to technical and physical limitations, an extent to which an area of a peripheral circuit is reduced may not reach an extent to which an area of the memory cell array is reduced.

[0005] In addition, as memory cell array chips are stacked, a distance between the memory cell array and the peripheral circuit may increase, which may increase load applied to the peripheral circuit. As a result, the increase in load may deteriorate the performance of the memory device.

SUMMARY

[0006] Embodiments of the present disclosure provide a memory device, which has a structure capable of improving the net die of semiconductor chips included in the memory device and may minimize load applied to the memory device when the memory device performs input/output (I/O) operations.

[0007] The technical aspects of embodiments of the present disclosure are not limited to those described above; other aspects may become apparent to those of ordinary skill in the art based on the following description.

[0008] According to embodiments of the present disclosure, a memory device may be provided and include: a first cell array chip including a first memory cell array; a first peripheral circuit chip including a first page buffer circuit electrically connected to the first memory cell array; an input/output chip including a first pad circuit, a second pad circuit, a first input/output circuit electrically connected to the first pad circuit, and a second input/output circuit electrically connected to the second pad circuit; a second cell array chip including a second memory cell array; and a second peripheral circuit chip including a second page buffer circuit electrically connected to the second memory cell array.

[0009] According to embodiments of the present disclosure, a memory device may be provided and include: a first cell array chip including a first memory cell array; a first peripheral circuit chip including a first page buffer circuit electrically connected to the first memory cell array; a second cell array chip including a second memory cell array; and a second peripheral circuit chip including a second page buffer circuit, a first pad circuit, a second pad circuit, a first input/output circuit electrically connected to the first pad circuit, and a second input/output circuit electrically

connected to the second pad circuit, wherein the second page buffer circuit is electrically connected to the second memory cell array and formed on one surface of the second peripheral circuit chip, and wherein the first pad circuit, the second pad circuit, the first input/output circuit, and the second input/output circuit are formed on another surface of the second peripheral circuit chip.
[0010] According to embodiments of the present disclosure, a memory device may be provided and include: a first cell array chip including a first memory cell array; a first peripheral circuit chip including a first page buffer circuit electrically connected to the first memory cell array; an input/output chip including a first pad circuit, a second pad circuit, a first input/output circuit electrically connected to the first pad circuit, and a second input/output circuit electrically connected to the second pad circuit; a second cell array chip including a second memory cell array; a second peripheral circuit chip including a second page buffer circuit electrically connected to the second memory cell array; a third cell array chip including a third memory cell array; and a third peripheral circuit chip including a third page buffer circuit electrically connected to the third memory cell array.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0011] Embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:
[0012] FIG. **1** is a block diagram of a storage device according to embodiments;
[0013] FIG. **2** is a block diagram of a memory device according to embodiments;
[0014] FIG. **3** is a view illustrating a structure of the memory device of FIG. **1**, according to embodiments;
[0015] FIGS. **4** and **5** are views illustrating memory devices according to embodiments;
[0016] FIGS. **6** and **7** are views illustrating memory devices according to embodiments;
[0017] FIG. **8** is a view illustrating a memory device according to embodiments;
[0018] FIGS. **9**A and **9**B are views illustrating an operating method of a memory device according to embodiments;
[0019] FIGS. **10**A and **10**B are views illustrating an operating method of a memory device according to embodiments;
[0020] FIGS. **11**A and **11**B are views illustrating an operating method of a memory device according to embodiments;
[0021] FIGS. **12**A and **12**B are views illustrating an operating method of a memory device according to embodiments;
[0022] FIGS. **13** to **16** are views respectively illustrating memory devices according to embodiments;
[0023] FIGS. **17** to **20** are views respectively illustrating memory devices according to embodiments;
[0024] FIG. **21** is a view illustrating a memory device according to embodiments;
[0025] FIG. **22** is a view illustrating a memory device according to embodiments;
[0026] FIG. **23** is a view illustrating a memory device according to embodiments;
[0027] FIG. **23** is a view illustrating a memory device according to embodiments;
[0028] FIG. **25** is a view illustrating a memory device according to embodiments; and
[0029] FIG. **26** is a block diagram of an example of applying a memory device according to embodiments to a solid-state drive (SSD) system.
DETAILED DESCRIPTION
[0030] Hereinafter, non-limiting example embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The same reference numerals are used to

denote the same elements in the drawings, and repeated descriptions thereof may be omitted.

[0031] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it can be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present.

[0032] FIG. **1** is a block diagram of a storage device **10** according to embodiments.

[0033] Referring to FIG. **1**, the storage device **10** may include storage mediums configured to store data in response to a request from a host. For example, the storage device **10** may include at least one from among a solid-state drive (SSD), an embedded memory, and a removable external memory. When the storage device **10** is the SSD, the storage device **10** may be a device that conforms to a non-volatile memory express (NVMe) standard.

[0034] When the storage device **10** is the embedded memory or the external memory, the storage device **10** may be a device that conforms to the standard of a universal flash storage (UFS) or an embedded multi-media card (eMMC). Each of the host and the storage device **10** may generate and transmit packets that conform to an adopted standard protocol. In an embodiment, the storage device **10** may be an embedded memory that is embedded in the storage device **10**. For example, the storage device **10** may include an eMMC or an embedded UFS memory device. In an embodiment, the storage device **10** may be external memory that is detachably attached to another system. For example, the storage device **10** may include a UFS memory card, a compact flash (CF), a secure digital (SD), a micro-SD, a mini-SD, an extreme Digital (xD), or a memory stick.

[0035] Referring to FIG. **1**, the storage device **10** may include a memory controller **100** and a memory device **200**.

[0036] The memory controller **100** may control read, write, and erase operations on the memory device **200** by providing an address ADDR, a command CMD, and a control signal CTRL to the memory device **200**. In this case, when the memory device **200** performs read and write operations, data DATA may be transmitted and received between the memory controller **100** and the memory device **200**.

[0037] In an embodiment, the memory controller **100** may control the memory device **200** to read data stored in the memory device **200** in response to a read request from a host HOST or write data to the memory device **200** in response to a write request from the host HOST. In an embodiment, the storage device **10** may perform operations, such as wear leveling management, bad block management, and garbage collection.

[0038] The memory device **200** may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, an input/output (I/O) chip IOC, a second peripheral circuit chip PC**2**, and a second cell array chip CAC**2**. The first cell array chip CAC**1**, the first peripheral circuit chip PC**1**, the I/O chip IOC, the second peripheral circuit chip PC**2**, and the second cell array chip CAC**2** may each be a semiconductor chip that is formed on a separate wafer and then diced. Each of the first cell array chip CAC**1** and the second cell array chip CAC**2** may include a memory cell array. The first peripheral circuit chip PC**1** may include circuits to control the memory cell array included in the first cell array chip CAC**1**. The second peripheral circuit chip PC**2** may include circuits to control the memory cell array included in the second cell array chip CAC**2**. The I/O chip IOC may include a pad circuit connected to a bonding pad and an I/O circuit to transmit and receive data to and from the first peripheral circuit chip PC**1** and the second peripheral circuit chip PC**2**. Component blocks included in each of the first cell array chip CAC**1**, the first peripheral circuit chip PC**1**, the I/O chip IOC, the second peripheral circuit chip PC**2**, and the second cell array chip CAC**2** are described below with reference to FIG. **2**.

[0039] FIG. **2** is a block diagram of a memory device **200** according to embodiments. FIG. **2** may be described with reference to FIG. **1**, and repeated descriptions may be omitted. Although only a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, and an I/O chip IOC are illustrated in

FIG. **2** for brevity, the memory device **200** of FIG. **2** may further include a second cell array chip CAC**2** and a second peripheral circuit chip PC**2**. In this case, the second cell array chip CAC**2** may have a similar configuration to the configuration of the first cell array chip CAC**1**, and the second peripheral circuit chip PC**2** may have a similar configuration to the configuration of the first peripheral circuit chip PC**1**.

[0040] Referring to FIG. **2**, the memory device **200** may include the first cell array chip CAC**1**, the first peripheral circuit chip PC**1**, and the I/O chip IOC.

[0041] The first peripheral circuit chip PC**1** may include a voltage generator **210**, an address decoder **220**, a control logic circuit **240**, and a page buffer circuit **250**. The first cell array chip CAC**1** may include a memory cell array **230**. The I/O chip PC**1** may include an I/O circuit **260**.

[0042] The voltage generator **210** may generate various kinds of voltages for performing read, write, and erase operations on the memory cell array **230**, based on a voltage control signal CTRL_Vol. Specifically, the voltage generator **210** may generate a word line voltage VWL, for example, a program voltage, a read voltage, a pass voltage, erase verification voltage, or a program verification voltage. Also, the voltage generator **210** may generate a string selection line voltage or a ground selection line voltage, based on the voltage control signal CTRL_Vol. In addition, the voltage generator **210** may generate an erase voltage to be provided to the memory cell array **230**.

[0043] The address decoder **220** may select one of a plurality of memory regions MR**1** to MRz (z is a natural number of 1 or more) of the memory cell array **230**, one of word lines WL of the selected memory region, and select one of a plurality of string selection lines SSL.

[0044] The memory cell array **230** may be connected to the word lines WL, the string selection lines SSL, ground selection lines GSL, and bit lines BL. The memory cell array **230** may be connected to the address decoder **220** through the word lines WL, the string selection lines SSL, and the ground selection lines GSL and be connected to the page buffer circuit **250** through the bit lines BL. The memory cell array **230** may include the plurality of memory regions MR**1** to MRz.

[0045] Each of the plurality of memory regions MR**1** to MRz may include a plurality of memory cells and a plurality of selection transistors. The memory cells may be connected to the word lines WL, and the selection transistors may be connected to the string selection lines SSL or the ground selection lines GSL. Each of the plurality of memory regions MR**1** to MRz may include a plurality of memory pages A memory page may correspond to a program unit of data or a read unit of data. As used herein, a memory region may refer to a group of a plurality of memory cells. Alternatively, each memory region may correspond to a memory block.

[0046] In an embodiment, the memory cell array **230** may include a three-dimensional (3D) a memory cell array, which includes a plurality of NAND strings. Each of the NAND strings may include memory cells respectively connected to word lines WL that are vertically stacked on a substrate. However, embodiments of the present disclosure are not limited thereto. In some embodiments, the memory cell array **230** may include a two-dimensional (2D) memory cell array, which includes a plurality of NAND strings arranged in a row direction and a column direction.

[0047] The control logic circuit **240** may output various control signals for performing read, write, and erase operations on the memory cell array **230**, based on a command CMD, an address ADDR, and a control signal CTRL. The control logic circuit **240** may provide a row address X-ADDR to the address decoder **220**, provide a column address Y-ADDR to the page buffer circuit **250**, and provide a voltage control signal CTRL_Vol to the voltage generator **210**.

[0048] The page buffer circuit **250** may include a plurality of page buffers PB**1** to PBn (n is a natural number of 1 or more). The plurality of page buffers PB**1** to PBn may respectively correspond to the plurality of memory regions MR**1** to MRz. The page buffer circuit **250** may operate as a write driver or a sense amplifier according to an operation mode. In a read operation, the page buffer circuit **250** may sense a bit line BL of a selected memory cell via the control of the control logic circuit **240**. Sensed data may be stored in latches included in the page buffer circuit **250**. The page buffer circuit **250** may dump the data stored in the latches to the I/O circuit **260** via the control of the control logic circuit **240**.

[0049] The I/O circuit **260** may temporarily store data DATA provided from the outside of the memory device **200**. The I/O circuit **260** may temporarily store read data of the memory device **200** and output data to the outside at a designated point in time. In addition, the I/O circuit **260** may temporarily store write data DATA provided from the outside and provide the data DATA to the page buffer circuit **250** at a designated point in time.

[0050] FIG. **3** is a view illustrating a structure of the memory device **200** of FIG. **1**, according to embodiments. FIG. **3** may be described with reference to FIGS. **1** and **2**, and repeated descriptions may be omitted.

[0051] FIG. **3** illustrates one example of structures of the memory device **200** of FIG. **1**. Referring to FIG. **3**, the memory device **200** may include first to fifth semiconductor layers L**1** to L**5**. The second semiconductor layer L**2** may be under the first semiconductor layer L**1** in a vertical direction (i.e., a first direction D**1**). The third semiconductor layer L**3** may be under the second semiconductor layer L**2** in the first direction D**1**. The fourth semiconductor layer L**4** may be under the third semiconductor layer L**3** in the first direction D**1**. The fifth semiconductor layer L**5** may be under the fourth semiconductor layer L**4** in the first direction D**1**.

[0052] As used herein, the vertical direction may be defined as the first direction D**1**, a second direction D**2** may be defined as a direction perpendicular to the first direction D**1**, and a third direction D**3** may be defined as a direction perpendicular to the first direction D**1** and the second direction D**2**. Also, the second direction D**2** may also be referred to as a first lateral direction, and the third direction D**3** may also be referred to as a second lateral direction.

[0053] In an embodiment, the memory device **200** may have a symmetrical structure about the third semiconductor layer L**3**. The first cell array chip CAC**1** of FIG. **1** may be formed in the first semiconductor layer L**1**. The first peripheral circuit chip PC**1** of FIG. **1** may be formed in the second semiconductor layer L**2**. The I/O chip IOC of FIG. **1** may be formed in the third semiconductor layer L**3**. The second cell array chip CAC**2** of FIG. **1** may be formed in the fourth semiconductor layer L**4**. The second peripheral circuit chip PC**2** of FIG. **1** may be formed in the fifth semiconductor layer L**5**.

[0054] The memory device **200** may include a chip-to-chip (C2C) structure. Here, the C2C structure may refer to a structure in which at least one upper chip including a cell region (e.g., a first cell region CELL**1**; refer to FIG. **25**) and a lower chip including a peripheral circuit region PERI (refer to FIG. **25**) are separately manufactured and connected to each other by using a bonding technique.

[0055] In an embodiment, the first cell array chip CAC**1** and the first peripheral circuit chip PC**1** may be connected to each other by using a bonding technique to have a C2C structure. In addition, the second cell array chip CAC**2** and the second peripheral circuit chip PC**2** may be connected to each other by using a bonding technique to have a C2C structure. A detailed description of the C2C structure is provided below with reference to FIG. **25**.

[0056] FIGS. **4** and **5** are views illustrating a memory device **200***a* and a memory device **200***b* according to embodiments. Specifically, each of FIGS. **4** and **5** corresponds to a cross-section of the memory device **200**, which is taken along line I-I′ of FIG. **3**. FIGS. **4** and **5** may be described with reference to FIGS. **1** to **3**, and repeated descriptions may be omitted.

[0057] Referring to FIG. **4**, the memory device **200***a* may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, an I/O chip IOC, a second peripheral circuit chip PC**2**, a second cell array chip CAC**2**, a first bonding pad PD**1**, a second bonding pad PD**2**, contact plugs PT**11**, PT**12**, PT**21**, and PT**22**, contact bonding pads PBP**11**, PBP**12**, PBP**21**, and PBP**22**, metal patterns MP**11**, MP**12**, MP**21**, and MP**22**, and first to fourth connection units CN**1***a* to CN**4***a*. Also, the first bonding pad PD**1** and the second bonding pad PD**2** may be located on the first cell array chip CAC**1**. The memory device **200***a* may communicate with the memory controller **100** (refer to FIG. **1**) through the first bonding pad PD**1** and the second bonding pad PD**2**. As used herein, a contact bonding pad may refer to a conductive pad located for electrical connection between a contact plug and another component. In this case, the other component electrically connected to the contact plug may be, for example, another contact plug or another semiconductor chip. The first peripheral circuit chip PC**1** may be electrically connected to the first cell array chip CAC**1** through the metal patterns MP**11** and MP**12** by using a bonding technique. The first peripheral circuit chip PC**1** may be electrically connected to the I/O chip IOC through the first connection unit CN**1***a* and the second connection unit CN**2***a*. The second peripheral circuit chip PC**2** may be electrically connected to the I/O chip IOC through the third connection unit CN**3***a* and the fourth connection unit CN**4***a*. The second peripheral circuit chip PC**2** may be electrically connected to the second cell array chip CAC**2** through the metal patterns MP**21** and MP**22** by using a bonding technique.

[0058] The first cell array chip CAC**1** may include a first memory cell array MCA**1**. The first memory cell array MCA**1** may include a first memory region MR**1** and a second memory region MR**2**. The first memory cell array MCA**1** is illustrated as including two memory regions in a descriptive sense only and not for purposes of limitation. That is, the first memory cell array MCA may include more than two memory regions.

[0059] The first peripheral circuit chip PC**1** may include a first page buffer circuit PBC**1**. The first page buffer circuit PBC**1** may include a first page buffer PB**1** and a second page buffer PB**2**. The first page buffer circuit PBC**1** is illustrated as including two page buffers in a descriptive sense only and not for purposes of limitation. That is, the first page buffer circuit PBC**1** may include more than two page buffers.

[0060] The second cell array chip CAC**2** may include a second memory cell array MCA**2**. The second memory cell array MCA**2** may include a third memory region MR**3** and a fourth memory region MR**4**. The second memory cell array MCA**2** is illustrated as including two memory regions in a descriptive sense only and not for purposes of limitation. That is, the second memory cell array MCA**2** may include more than two memory regions.

[0061] The second peripheral circuit chip PC**2** may include a second page buffer circuit PBC**2**. The second page buffer circuit PBC**2** may include a third page buffer PB**3** and a fourth page buffer PB**4**. The second page buffer circuit PBC**2** is illustrated as including two page buffers in a descriptive sense only and not for purposes of limitation. That is, the second page buffer circuit PBC**2** may include more than two page buffers.

[0062] The I/O chip IOC may include a first I/O circuit **301**, a second I/O circuit **302**, a first pad circuit **303**, and a second pad circuit **304**. Each of the first I/O circuit **301** and the second I/O circuit **302** may correspond to the I/O circuit **260** of FIG. **2**. The first I/O circuit **301** may be apart from the second I/O circuit **302** in the third direction D**3**. The first pad circuit **303** may be closer to the first I/O circuit **301** than the second I/O circuit **302**. The second pad circuit **304** may be closer to the second I/O circuit **302** than the first I/O circuit **301**. The first memory region MR**1**, the first page buffer PB**1**, the third page buffer PB**3**, and the third memory region MR**3** may be closer to the first I/O circuit **301** than the second I/O circuit **302**. The second memory region MR**2**, the second page buffer PB**2**, the fourth page buffer PB**4**, and the fourth memory region MR**4** may be closer to the second I/O circuit **302** than the first I/O circuit **301**.

[0063] The I/O chip IOC may be divided into a first I/O unit IOC_A**1** and a second I/O unit IOC_A**2**. As used herein, the first I/O unit IOC_A**1** may refer to a region including the first I/O circuit **301** and the first pad circuit **303**. The second I/O unit IOC_A**2** may refer to a region including the second I/O circuit **302** and the second pad circuit **304**.

[0064] The first pad circuit **303** may be a circuit configured to electrically connect the first I/O circuit **301** to the first bonding pad PD**1**. The second pad circuit **304** may be a circuit configured to electrically connect the second I/O circuit **302** to the second bonding pad PD**2**. The first bonding pad PD**1** may be electrically connected to the first pad circuit **303** through the contact plug PT**11** and the contact plug PT**21**. The contact plug PT**11** and the contact plug PT**12** may pass through the first cell array chip CAC**1**. The contact plug PT**21** and the contact plug PT**22** may pass through the first peripheral circuit chip PC**1**. The contact plug PT**11** may be electrically connected to the contact plug PT**21** through the contact bonding pad PBP**11**. The contact plug PT**21** may be electrically connected to the I/O chip IOC through the contact bonding pad PBP**21**. The contact plug PT**12** may be electrically connected to the contact plug PT**22** through the contact bonding pad PBP**12**. The contact plug PT**22** may be electrically connected to the I/O chip IOC through the contact bonding pad PBP**22**. In an embodiment, the contact bonding pads PBP**11** and PBP**12** may be between the first cell array chip CAC**1** and the first peripheral circuit chip PC**1**. In an embodiment, the contact bonding pads PBP**21** and PBP**22** may be between the first peripheral circuit chip PC**1** and the I/O chip IOC.

[0065] According to embodiments, a first contact passing through the first peripheral circuit chip PC**1** and the first cell array chip CAC**1** may be provided. For example, the first contact may include the contact plug PT**11** and the contact plug PT**21**. According to embodiments, a second contact passing through the first peripheral circuit chip PC**1** and the first cell array chip CAC**1** may be provided. The second contact may include the contact plug PT**12** and the contact plug PT**21**.

[0066] The memory device **200***a* may perform an I/O operation (e.g., a write operation or a read operation) related to data DATA, by using an I/O circuit that is close to each memory region.

[0067] In an embodiment, the memory device **200***a* may write data DATA to the first cell array chip CAC**1** or read the data DATA from the first cell array chip CAC**1** through a first path PATH**1**. The first path PATH**1** may refer to a path passing through the first cell array chip CAC**1**, the first peripheral circuit chip PC**1**, and the I/O chip IOC. In this case, the memory device **200***a* may process the input and output of data related to the first memory region MR**1** by using the first page buffer PB**1** and the first I/O circuit **301**, and process the input and output of data related to the second memory region MR**2** by using the second page buffer PB**2** and the second I/O circuit **302**.

[0068] In an embodiment, the memory device **200***a* may write data DATA to the second cell array chip CAC**2** or read the data DATA from the second cell array chip CAC**2** through a second path PATH**2**. The second path PATH**2** may refer to a path passing through the second cell array chip CAC**2**, the second peripheral circuit chip PC**2**, and the I/O chip IOC. In this case, the memory device **200***a* may process the input and output of data related to the third memory region MR**3** by using the third page buffer PB**3** and the first I/O circuit **301**, and process the input and output of data related to the fourth memory region MR**4** by using the fourth page buffer PB**4** and the second I/O circuit **302**.

[0069] The memory device **200***a* may perform 1-channel communication with the memory controller **100**. That is, the memory device **200***a* may transmit and receive data DATA to and from the memory controller **100** through the first bonding pad PD**1** and the second bonding pad PD**2** in response to a first command received from the memory controller **100**. The data DATA may include first data DATA**1** and second data DATA**2**. The first data DATA**1** may correspond to lower bits, from among bits that constitute the data DATA, and the second data DATA**2** may correspond to upper bits, from among the bits that constitute the data DATA. However, embodiments of the present disclosure are not limited thereto, and the first data DATA**1** may correspond to the upper bits, from among bits that constitute the data DATA, and the second data DATA**2** may correspond to the lower bits, from among the bits that constitute the data DATA.

[0070] In an embodiment, a size unit of the data DATA that are input and output between the memory device **200***a* and the memory controller **100** may be 16 bits. In this case, the memory device **200***a* may operate as a wide I/O interface. The first data DATA**1** may correspond to lower 8-bit data of the data DATA, and the second data DATA**2** may correspond to upper 8-bit data of the data DATA.

[0071] In an embodiment, a size unit of the data DATA that are input and output between the memory device **200***a* and the memory controller **100** may be 8 bits. In this case, the first data DATA**1** may correspond to lower 4-bit data of the data DATA, and the second data DATA**2** may correspond to upper 4-bit data of the data DATA.

[0072] Referring to FIG. **5**, the memory device **200***b* may include components corresponding to the memory device **200***a* of FIG. **4**. Hereinafter, repeated descriptions may be omitted, and the description below focuses on differences between the memory device **200***b* of FIG. **5** and the memory device **200***a* of FIG. **4**.

[0073] However, unlike in the memory device **200***a* of FIG. **4**, a first peripheral circuit chip PC**1** of the memory device **200***b* of FIG. **5** may be electrically connected to a first I/O unit IO_A**1** through a first connection unit CN**1***b*. Similarly, the second peripheral circuit chip PC**2** of the memory device **200***b* of FIG. **5** may be electrically connected to the second I/O unit IO_A**2** through a second connection unit CN**2***b.*

[0074] The memory device **200***b* may process an I/O operation related to a first cell array chip CAC**1** by using a first I/O circuit **301** and perform an I/O operation related to a second cell array chip CAC**2** by using the second I/O circuit **302**.

[0075] In an embodiment, the memory device **200***b* may write first data DATA**1** to the first cell array chip CAC**1** or read the first data DATA**1** from the first cell array chip CAC**1** through the first path PATH**1**. The first path PATH**1** may refer to a path passing through the first cell array chip CAC**1**, the first peripheral circuit chip PC**1**, and the first I/O unit IOC_A**1**. In this case, the memory device **200***b* may process the input and output of the first data DATA**1** related to the first memory cell array MCA**1** by using the first page buffer circuit PBC**1** and the first I/O circuit **301**.

[0076] In an embodiment, the memory device **200***b* may write second data DATA**2** to the second cell array chip CAC**2** or read the second data DATA**2** from the second cell array chip CAC**2** through the second path PATH**2**. The second path PATH**2** may refer to a path passing through the second cell array chip CAC**2**, the second peripheral circuit chip PC**2**, and the second I/O unit IOC_A**2**. In this case, the memory device **200***b* may process the input and output of the second data DATA**2** related to the second memory cell array MCA**2** by using the second page buffer circuit PBC**2** and the second I/O circuit **302**.

[0077] The memory device **200***b* may perform 2-channel communication with a memory controller **100**. The memory device **200***b* may transmit and receive the first data DATA**1** to and from the memory controller **100** through the first bonding pad PD**1** in response to a first command received from the memory controller **100**. The memory device **200***b* may transmit and receive the second data DATA**2** to and from the memory controller **100** through the second bonding pad PD**2** in response to a second command received from the memory controller **100**. In this case, the second command may be a different command from the first command.

[0078] In an embodiment, a size unit of the first data DATA**1** and the second data DATA**2** that are input and output between the memory device **200***b* and the memory controller **100** may be 16 bits. In this case, the memory device **200***b* may operate as a wide I/O interface. The first data DATA**1** may correspond to 16-bit data, and the second data DATA**2** may also correspond to 16-bit data.

[0079] In an embodiment, a size unit of the first data DATA**1** and the second data DATA**2** that are input and output between the memory device **200***b* and the memory controller **100** may be 8 bits. In this case, the first data DATA**1** may correspond to 8-bit data, and the second data DATA**2** may also correspond to 8-bit data.

[0080] FIGS. **6** and **7** are views illustrating a memory device **200***c* and a memory device **200***d* according to embodiments. Specifically, each of FIGS. **6** and **7** corresponds to a cross-section of the memory device **200**, which is taken along line I-I′ of FIG. **3**. FIGS. **6** and **7** may be described with reference to FIGS. **4** and **5**, and repeated descriptions may be omitted.

[0081] Referring to FIG. **6**, the memory device **200***c* may include a component corresponding to the memory device **200***a* of FIG. **4**. Hereinafter, repeated descriptions may be omitted, and the description below focuses on differences between the memory device **200***c* of FIG. **6** and the memory device **200***a* of FIG. **4**.

[0082] The memory device **200***c* of FIG. **6** may include a first edge pad EPD**1** and a second edge pad EPD**2**. The first edge pad EPD**1** may be a component corresponding to the first bonding pad

PD**1** of the memory device **200***a* of FIG. **4**. The second edge pad EPD**2** may be a component corresponding to the second bonding pad PD**2** of the memory device **200***a* of FIG. **4**. The first edge pad EPD**1** may be electrically connected to a first pad circuit **303**. The second edge pad EPD**2** may be electrically connected to the second pad circuit **304**.

[0083] The first edge pad EPD**1** and the second edge pad EPD**2** may be apart from each other in a third direction D**3**. The first edge pad EPD**1** and the second edge pad EPD**2** may be between a first peripheral circuit chip PC**1** and an I/O chip IOC. However, embodiments of the present disclosure are not limited thereto, and the first edge pad EPD**1** and the second edge pad EPD**2** may be disposed between a second peripheral circuit chip PC**2** and the I/O chip IOC.

[0084] In an embodiment, the first edge pad EPD**1** and the second edge pad EPD**2** may be electrically connected to a memory controller **100** by using a wire bonding process.

[0085] In an embodiment, the first edge pad EPD**1** and the second edge pad EPD**2** may be electrically connected to the memory controller **100** through a film cable.

[0086] In an embodiment, the first edge pad EPD**1** and the second edge pad EPD**2** may be electrically connected to the memory controller **100** through a conductive epoxy material.

[0087] Referring to FIG. **7**, the memory device **200***d* may include a component corresponding to the memory device **200***b* of FIG. **5**. Hereinafter, repeated descriptions may be omitted, and the description below focuses on differences between the memory device **200***d* of FIG. **7** and the memory device **200***b* of FIG. **5**.

[0088] The memory device **200***d* of FIG. **7** may include a first edge pad EPD**1** and a second edge pad EPD**2**. The first edge pad EPD**1** may be a component corresponding to the first bonding pad PD**1** of the memory device **200***b* of FIG. **5**. The second edge pad EPD**2** may include a component corresponding to the second bonding pad PD**2** of the memory device **200***b* of FIG. **5**. In the memory device **200***d* of FIG. **7**, the first edge pad EPD**1** and the second edge pad EPD**2** may be arranged at the same positions as in FIG. **6**, and thus, repeated descriptions may be omitted.

[0089] FIG. **8** is a view illustrating a memory device **200***e* according to embodiments. FIG. **8** may be described with reference to FIGS. **6** and **7**, and repeated descriptions may be omitted.

[0090] Referring to FIG. **8**, the memory device **200***e* may be obtained by stacking a plurality of the memory device **200***d* of FIG. **7** in two stages. However, embodiments of the present disclosure are not limited to an example in which a plurality of the memory device **200***d* of FIG. **7** are stacked in two stages. In another case, a plurality of the memory device **200***c* of FIG. **6** may be stacked in two stages. Also, at least two memory devices may be stacked.

[0091] The memory device **200***e* of FIG. **8** may include a first memory device **200***e*_**1** and a second memory device **200***e*_**2**. The first memory device **200***e*_**1** may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, a first I/O chip IOC**1**, a second peripheral circuit chip PC**2**, and a second cell array chip CAC**2**. The second memory device **200***e*_**2** may include a third cell array chip CAC**3**, a third peripheral circuit chip PC**3**, a second I/O chip IOC**2**, a fourth peripheral circuit chip PC**4**, and a fourth cell array chip CAC**4**.

[0092] The first peripheral circuit chip PC**1** may be under the first cell array chip CAC**1**. The first I/O chip IOC**1** may be under the first peripheral circuit chip PC**1**. The second peripheral circuit chip PC**2** may be under the first I/O chip IOC**1**. The second cell array chip CAC**2** may be under the second peripheral circuit chip PC**2**. The third cell array chip CAC**3** may be under the second cell array chip CAC**2**. The third peripheral circuit chip PC**3** may be under the third cell array chip CAC**3**. The second I/O chip IOC**2** may be under the third peripheral circuit chip PC**3**. The fourth peripheral circuit chip PC**4** may be under the second I/O chip IOC**2**. The fourth cell array chip CAC**4** may be under the fourth peripheral circuit chip PC**4**.

[0093] The first cell array chip CAC**1** and the first peripheral circuit chip PC**1** may be bonded to each other through the metal patterns MP**11** and MP**12**. The second cell array chip CAC**2** and the second peripheral circuit chip PC**2** may be bonded to each other through the metal patterns MP**21** and MP**22**. The third cell array chip CAC**3** and the third peripheral circuit chip PC**3** may be bonded to each other through the metal patterns MP**31** and MP**32**. The fourth cell array chip CAC**4** and the fourth peripheral circuit chip PC**4** may be bonded to each other through the metal patterns MP**41** and MP**42**.

[0094] The first peripheral circuit chip PC**1** and the first I/O chip IOC**1** may be electrically connected to each other through a first connection unit CN**1***e*. The second peripheral circuit chip PC**2** and the first I/O chip IOC**1** may be electrically connected to each other through a second connection unit CN**2***e*. The third peripheral circuit chip PC**3** and the second I/O chip IOC**2** may be electrically connected to each other through a third connection unit CN**3***e*. The fourth peripheral circuit chip PC**4** and the second I/O chip IOC**2** may be electrically connected to each other through a fourth connection unit CN**4***e*. However, the second cell array chip CAC**2** and the third cell array chip CAC**3** may not be electrically connected to each other, and may only be physically connected to each other by adhesives or insulating materials.

[0095] The first memory device **200***e*_**1** may provide first data DATA**1** and second data DATA**2** to the memory controller **100** or receive the first data DATA**1** and the second data DATA**2** from the memory controller **100** in response to a command received from a memory controller **100**.

[0096] The second memory device **200***e*_**2** may provide third data DATA**3** and fourth data DATA**4** to the memory controller **100** or receive the third data DATA**3** and the fourth data DATA**4** from the memory controller **100** in response to a command received from the memory controller **100**.

[0097] In an embodiment, a first edge pad EPD**1** may be electrically connected to a third edge pad EPD**3** by using a wire bonding process. A second edge pad EPD**2** may be electrically connected to a fourth edge pad EPD**4** by using a wire bonding process. The third edge pad EPD**3** and the fourth edge pad EPD**4** may be electrically connected to an external device by using a wire bonding process.

[0098] In an embodiment, the first edge pad EPD**1** may be electrically connected to the third edge pad EPD**3** through a film cable. The second edge pad EPD**2** may be electrically connected to the fourth edge pad EPD**4** through a film cable. The third edge pad EPD**3** and the fourth edge pad EPD**4** may be electrically connected to an external device through a film cable.

[0099] In an embodiment, the first edge pad EPD**1** may be electrically connected to the third edge pad EPD**3** through a conductive epoxy material. The second edge pad EPD**2** may be electrically connected to the fourth edge pad EPD**4** through a conductive epoxy material. The third edge pad EPD**3** and the fourth edge pad EPD**4** may be electrically connected to an external device through a conductive epoxy material.

[0100] FIGS. **9**A and **9**B are views illustrating an operating method of a memory device **200** according to embodiments. FIGS. **9**A and **9**B may be described with reference to FIGS. **1** to **5**, and repeated descriptions may be omitted.

[0101] Referring to FIGS. **9**A and **9**B, the memory device **200** may include a first memory cell array MCA**1**, a first page buffer circuit PBC**1**, a first I/O unit IOC_A**1***a*, a second I/O unit IOC_A**2***a*, a second page buffer circuit PBC**2**, and a second memory cell array MCA**2**.

[0102] The first memory cell array MCA**1** may be formed in the first cell array chip CAC**1** of FIG. **1**. The first memory cell array MCA**1** may include first to eighth memory regions MR**1***u* to MR**8***u*. Although the first memory cell array MCA**1** is illustrated as including eight memory regions as an example, the first memory cell array MCA**1** may include fewer or more memory regions than in FIGS. **9**A and **9**B.

[0103] The first page buffer circuit PBC**1** may be formed in the first peripheral circuit chip PC**1** of FIG. **1**. The first page buffer circuit PBC**1** may include first to eighth page buffers PB**1***u* to PB**8***u*. Although the first page buffer circuit PBC**1** is illustrated as including eight page buffers as an example, the first page buffer circuit PBC**1** may include fewer or more page buffers than in FIGS. **9**A and **9**B.

[0104] In an embodiment, the first to eighth page buffers PB**1***u* to PB**8***u* may respectively overlap with the first to eighth memory regions MR**1***u* to MR**8***u* in a first direction D**1**. For example, the first memory region MR**1***u* and the first page buffer PB**1***u* may overlap each other in the first direction D**1**. For example, the sixth memory region MR**6***u* and the sixth page buffer PB**6***u* may overlap each other in the first direction D**1**.

[0105] The first I/O unit IOC_A**1***a* and the second I/O unit IOC_A**2***a* may be formed in the I/O chip IOC of FIG. **1**. The first I/O unit IOC_A**1***a* may include a first I/O circuit **301** and a first pad circuit **303**. The second I/O unit IOC_A**2***a* may include a second I/O circuit **302** and a second pad circuit **304**. The first I/O unit IOC_A**1***a* and the second I/O unit IOC_A**2***a* may be apart from each other in a second direction D**2**. The first I/O circuit **301** and the first pad circuit **303** may be apart from each other in the second direction D**2** and may be electrically connected to each other through a wiring. The second I/O circuit **302** and the second pad circuit **304** may be apart from each other in the second direction D**2** and may be electrically connected to each other through a wiring.

[0106] The second page buffer circuit PBC**2** may be formed in the second peripheral circuit chip PC**2** of FIG. **1**. The second page buffer circuit PBC**2** may include first to eighth page buffers PB**1***d* to PB**8***d*. Although the second page buffer circuit PBC**2** is illustrated as including eight page buffers as an example, the second page buffer circuit PBC**2** may include fewer or more page buffers than in FIGS. **9**A and **9**B.

[0107] The second memory cell array MCA**2** may be formed in the second cell array chip CAC**2** of FIG. **1**. The second memory cell array MCA**2** may include first to eighth memory regions MR**1***d* to MR**8***d*. Although the second memory cell array MCA**2** is illustrated as including eight memory regions as an example, the second memory cell array MCA**2** may include fewer or more memory regions than in FIGS. **9**A and **9**B.

[0108] In an embodiment, the first to eighth page buffers PB**1***d* to PB**8***d* may respectively overlap with the first to eighth memory regions MR**1***d* to MR**8***d* in the first direction D**1**. For example, the fourth memory region MR**4***d* and the fourth page buffer PB**4***d* may overlap each other in the first direction D**1**. For example, the fifth memory region MR**5***d* and the fifth page buffer PB**5***d* may overlap each other in the first direction D**1**.

[0109] In an embodiment, the first page buffer circuit PBC**1** may be formed on one surface of the first peripheral circuit chip PC**1** of FIG. **1**, and the first I/O unit IOC_A**1***a* and the second I/O unit IOC_A**2***a* may be formed on another surface of the first peripheral circuit chip PC**1** of FIG. **1**.

[0110] In an embodiment, the second page buffer circuit PBC**2** may be formed on one surface of the second peripheral circuit chip PC**2** of FIG. **1**, and the first I/O unit IOC_A**1***a* and the second I/O unit IOC_A**2***a* may be formed on another surface of the second peripheral circuit chip PC**2** of FIG. **1**.

[0111] When reading data stored in the first memory cell array MCA**1** or when writing data to the first memory cell array MCA**1**, the memory device **200** may perform an operation of reading or writing data by using the first page buffer circuit PBC**1** and the first I/O unit IOC_A**1***a*.

[0112] In an embodiment, referring to FIG. **9**A, the memory device **200** may read data from the first memory region MR**1***u* or write data to the first memory region MR**1***u* through the first page buffer PB**1***u* and the first I/O circuit **301**.

[0113] In an embodiment, referring to FIG. **9**B, the memory device **200** may read data from the sixth memory region MR**6***u* or write data to the sixth memory region MR**6***u* through the sixth page buffer PB**6***u* and the first I/O circuit **301**.

[0114] When reading data stored in the second memory cell array MCA**2** or writing data to the second memory cell array MCA**2**, the memory device **200** may perform an operation of reading or writing data by using the second page buffer circuit PBC**2** and the second I/O unit IOC_A**2***a*.

[0115] In an embodiment, referring to FIG. **9**A, the memory device **200** may read data from the fifth memory region MR**5***d* or write data to the fifth memory region MR**5***d* through the fifth page buffer PB**5***d* and the second I/O circuit **302**.

[0116] In an embodiment, referring to FIG. **9**B, the memory device **200** may read data from the fourth memory region MR**4***d* or write data to the fourth memory region MR**4***d* through the fourth

page buffer PB**4***d* and the second I/O circuit **302**.
[0117] FIGS. **10**A and **10**B are views illustrating an operating method of a memory device **200** according to embodiments. FIGS. **10**A and **10**B may be described with reference to FIGS. **1** to **5**, **9**A, and **9**B, and repeated descriptions may be omitted.
[0118] The description below focuses on differences between operations of the memory device **200** shown in FIGS. **10**A and **10**B and the operations of the memory device **200** shown in FIGS. **9**A and **9**B.
[0119] Referring to FIGS. **10**A and **10**B, when reading or writing data from or to memory regions included a first memory cell array MCA**1**, the memory device **200** may perform an operation of reading or writing data by using an I/O circuit that is close to each memory region.
[0120] In an embodiment, referring to FIG. **10**A, the memory device **200** may read data from a third memory region MR**3***u* or write data to the third memory region MR**3***u* through a third page buffer PB**3***u* and a first I/O circuit **301**. In this case, a distance between the third memory region MR**3***u* and the first I/O circuit **301** may be less than a distance between the third memory region MR**3***u* and a second I/O circuit **302**.
[0121] In an embodiment, referring to FIG. **10**B, the memory device **200** may read data from a sixth memory region MR**6***u* or write data to the sixth memory region MR**6***u* through a sixth page buffer PB**6***u* and a second I/O circuit **302**. In this case, a distance between the sixth memory region MR**6***u* and the second I/O circuit **302** may be less than a distance between the sixth memory region MR**6***u* and the first I/O circuit **301**.
[0122] When reading or writing data from or to memory regions included in a second memory cell array MCA**2**, the memory device **200** may perform an operation of reading or writing data through an I/O circuit that is close to each memory region.
[0123] In an embodiment, referring to FIG. **10**A, the memory device **200** may read data from a sixth memory region MR**6***d* or write data to the sixth memory region MR**6***d* through a sixth page buffer PB**6***d* and the second I/O circuit **302**. In this case, a distance between the sixth memory region MR**6***d* and the second I/O circuit **302** may be less than a distance between the sixth memory region MR**6***d* and the first I/O circuit **301**.
[0124] In an embodiment, referring to FIG. **10**B, the memory device **200** may read data from a fourth memory region MR**4***d* or write data to the fourth memory region MR**4***d* through a fourth page buffer PB**4***d* and the first I/O circuit **301**. In this case, a distance between the fourth memory region MR**4***d* and the first I/O circuit **301** may be less than a distance between the fourth memory region MR**4***d* and the second I/O circuit **302**.
[0125] FIGS. **11**A and **11**B are views illustrating an operating method of a memory device **200** according to embodiments. FIGS. **11**A and **11**B may be described with reference to FIGS. **1** to **5**, **9**A, and **9**B, and repeated descriptions may be omitted.
[0126] The description below focuses on differences between operations of the memory device **200** shown in FIGS. **11**A and **11**B and the operations of the memory device **200** shown in FIGS. **9**A and **9**B.
[0127] Referring to FIGS. **11**A and **11**B, unlike the first I/O unit IOC_A**1***a* and the second I/O unit IOC_A**2***a* of FIGS. **9**A and **9**B, a first I/O unit IOC_A**1***b* and a second I/O unit IOC_A**2***b* of FIGS. **11**A and **11**B may be apart from each other in a third direction D**3**. The first pad circuit **303** and the second pad circuit **304** may be apart from each other in the third direction D**3**. A first I/O circuit **301** and the first pad circuit **303** may be apart from each other in a second direction D**2** and may be electrically connected to each other through a wiring. A second I/O circuit **302** and the second pad circuit **304** may be apart from each other in the second direction D**2** and may be electrically connected to each other through a wiring.
[0128] When reading data stored in a first memory cell array MCA**1** or when writing data to the first memory cell array MCA**1**, the memory device **200** may perform an operation of reading or writing data by using a first page buffer circuit PBC**1** and the first I/O unit IOC_A**1***a*.

[0129] In an embodiment, referring to FIG. **11**A, the memory device **200** may read data from a first memory region MR**1***u* or write data to the first memory region MR**1***u* through a first page buffer PB**1***u* and the first I/O circuit **301**.
[0130] In an embodiment, referring to FIG. **11**B, the memory device **200** may read data from an eighth memory region MR**8***u* or write data to the eighth memory region MR**8***u* through an eighth page buffer PB**8***u* and the first I/O circuit **301**.
[0131] When reading data stored in a second memory cell array MCA**2** or when writing data to the second memory cell array MCA**2**, the memory device **200** may perform an operation of reading or write data by using a second page buffer circuit PBC**2** and the second I/O unit IOC_A**2***a*.
[0132] In an embodiment, referring to FIG. **11**A, the memory device **200** may read data from a seventh memory region MR**7***d* or write data to the seventh memory region MR**7***d* through a seventh page buffer PB**7***s* and the second I/O circuit **302**.
[0133] In an embodiment, referring to FIG. **11**B, the memory device **200** may read data from a second memory region MR**2***d* or write data to the second memory region MR**2***d* through the second page buffer PB**2***d* and the second I/O circuit **302**.
[0134] FIGS. **12**A and **12**B are views illustrating an operating method of a memory device **200** according to embodiments. FIGS. **12**A and **12**B may be described with reference to FIGS. **1** to **5**, **11**A, and **11**B, and repeated descriptions may be omitted.
[0135] The description below focuses on differences between operations of the memory device **200** shown in FIGS. **12**A and **12**B and the operations of the memory device **200** shown in FIGS. **11**A and **11**B.
[0136] Referring to FIGS. **12**A and **12**B, when reading or writing data from or to memory regions included in a first memory cell array MCA**1**, the memory device **200** may perform an operation of reading or writing data by using an I/O circuit that is close to each of the memory regions.
[0137] In an embodiment, referring to FIG. **12**A, the memory device **200** may read data from a fifth memory region MR**5***u* or write data to the fifth memory region MR**5***u* through a fifth page buffer PB**5***u* and a first I/O circuit **301**. In this case, a distance between the fifth memory region MR**5***u* and the first I/O circuit **301** may be less than a distance between the fifth memory region MR**5***u* and a second I/O circuit **302**.
[0138] In an embodiment, referring to FIG. **12**B, the memory device **200** may read data from an eighth memory region MR**8***u* or write data to the eighth memory region MR**8***u* through an eighth page buffer PB**8***u* and the second I/O circuit **302**. In this case, a distance between the eighth memory region MR**8***u* and the second I/O circuit **302** may be less than a distance between the eighth memory region MR**8***u* and the first I/O circuit **301**.
[0139] When reading or writing data from or to memory regions included in a second memory cell array MCA**2**, the memory device **200** may perform an operation of reading or writing data by using an I/O circuit that is close to each of the memory regions.
[0140] In an embodiment, referring to FIG. **12**A, the memory device **200** may read data from a fourth memory region MR**4***d* or write data to the fourth memory region MR**4***d* through a fourth page buffer PB**4***d* and the second I/O circuit **302**. In this case, a distance between the fourth memory region MR**4***d* and the second I/O circuit **302** may be less than a distance between the fourth memory region MR**4***d* and the first I/O circuit **301**.
[0141] In an embodiment, referring to FIG. **12**B, the memory device **200** may read data from a second memory region MR**2***d* or write data to the second memory region MR**2***d* through a second page buffer PB**2***d* and the first I/O circuit **301**. In this case, a distance between the second memory region MR**2***d* and the first I/O circuit **301** may be less than a distance between the second memory region MR**2***d* and the second I/O circuit **302**.
[0142] FIGS. **13** to **16** are respectively views illustrating the memory device **200***a*, a memory device **200***b*, the memory device **200***c*, and the memory device **200***d* according to embodiments. Specifically, FIG. **13** is a view illustrating connection between chips included in the memory device

200*a* of FIG. **4**. FIG. **14** is a view illustrating connection between chips included in the memory device **200***b* of FIG. **5**. FIG. **15** is a view illustrating connection between chips included in the memory device **200***c* of FIG. **6**. FIG. **16** is a view illustrating connection between chips included in the memory device **200***d* of FIG. **7**. FIGS. **13** to **16** may be described with reference to FIGS. **4** to **7**, and repeated descriptions may be omitted.

[0143] Referring to FIGS. **13** and **15**, a first peripheral circuit chip PC**1** may be electrically connected to an I/O chip IOC through a first connection unit CN**1***a* and a second connection unit CN**2***a*. The first connection unit CN**1***a* may include a through-silicon via (TSV) TSV**11** and a TSV connection unit BP**11**. the second connection unit CN**2***a* may include a TSV TSV**12** and a TSV connection unit BP**12**. A second peripheral circuit chip PC**2** may be electrically connected to the I/O chip IOC through a third connection unit CN**3***a* and a fourth connection unit CN**4***a*. The third connection unit CN**3***a* may include a TSV TSV**21**, a TSV connection unit BP**21**, and a TSV connection unit BP**31**. The fourth connection unit CN**4***a* may include a TSV TSV**22**, a TSV connection unit BP**22**, and a TSV connection unit BP**32**.

[0144] The TSV TSV**11** may pass through the first peripheral circuit chip PC**1** and be electrically connected to the TSV connection unit BP**11**. The TSV connection unit BP**11** and the TSV connection unit BP**21** may be electrically connected to a first I/O circuit **301**. The TSV TSV**21** may pass through the I/O chip IOC and be electrically connected to the TSV connection unit BP**21** and the TSV connection unit BP**31**. A TSV TSV**31** may pass through the second peripheral circuit chip PC**2** and electrically connect the TSV connection unit BP**31** to components included in the second peripheral circuit chip PC**2**. The TSV TSV**12** may pass through the first peripheral circuit chip PC**1** and be electrically connected to the TSV connection unit BP**12**. The TSV connection unit BP**12** and the TSV connection unit BP**22** may be electrically connected to a second I/O circuit **302**. The TSV TSV**22** may pass through the I/O chip IOC and be electrically connected to the TSV connection unit BP**22** and the TSV connection unit BP**32**. A TSV TSV**32** may pass through the second peripheral circuit chip PC**2** and electrically connect the TSV connection unit BP**32** to components included in the second peripheral circuit chip PC**2**.

[0145] Referring to FIGS. **14** and **16**, the first peripheral circuit chip PC**1** may be electrically connected to the I/O chip IOC through a first connection unit CN**1***b*. The first connection unit CN**1***b* may include a TSV TSV**11** and a TSV connection unit BP**11**. The TSV TSV**11** may pass through the first peripheral circuit chip PC**1** and be electrically connected to the TSV connection unit BP**11**. The TSV connection unit BP**11** may be electrically connected to a TSV TSV**61** (e.g., refer to FIGS. **21-22**) through the first I/O circuit **301**.

[0146] The second peripheral circuit chip PC**2** may be electrically connected to the I/O chip IOC through a second connection unit CN**2***b*. The second connection unit CN**2***b* may include a TSV TSV**22**, a TSV connection unit BP**22**, and a TSV connection unit BP**32**. The TSV TSV**22** may pass through the I/O chip IOC and be electrically connected to the TSV connection unit BP**22**. The TSV connection unit BP**22** may be electrically connected to the TSV TSV**61** (e.g., refer to FIGS. **21-22**) through the second I/O circuit **302**. A TSV TSV**32** may pass through the second peripheral circuit chip PC**2** and electrically connect components included in the second peripheral circuit chip PC**2** to the TSV connection unit BP**32**.

[0147] Referring to FIGS. **13** and **14**, the contact plug PT**21** may be electrically connected to the first pad circuit **303** through the contact bonding pad PBP**21**. The contact plug PT**22** may be electrically connected to the second pad circuit **304** through the contact bonding pad PBP**22**.

[0148] As used herein, a TSV connection unit may refer to a component provided for electrically connection between a TSV and another component.

[0149] In an embodiment, the TSV connection units BP**11**, BP**12**, BP**21**, BP**22**, BP**31**, and BP**32** may include conductive pads. The conductive pads may include copper (Cu), aluminum (Al), tungsten (W), titanium (Ti), tantalum (Ta), indium (In), molybdenum (Mo), manganese (Mn), cobalt (Co), tin (Sn), nickel (Ni), magnesium (Mg), rhenium (Re), beryllium (Be), gallium (Ga),

and ruthenium (Ru), or an alloy thereof.
[0150] In an embodiment, the TSV connection units BP**11**, BP**12**, BP**21**, BP**22**, BP**31**, and BP**32** may include conductive bumps. The conductive bumps may include a metal, such as tin (Sn), copper (Cu), silver (Ag), gold (Au), tungsten (W), bismuth (Bi), zirconium (Zr), zinc (Zn), cobalt (Co), and nickel (Ni), or an alloy thereof.
[0151] FIGS. **17** to **20** are views respectively illustrating a memory device **200***f*, a memory device **200***g*, a memory device **200***h*, and a memory device **200***i* according to embodiments. FIGS. **17** to **20** may be described with reference to FIGS. **13** to **16**, and repeated descriptions may be omitted.
[0152] Referring to FIG. **17**, the memory device **200***f* of FIG. **17** may correspond to the memory device **200***a* of FIG. **13**. However, the memory device **200***f* of FIG. **17** may not include the I/O chip IOC unlike the memory device **200***a* of FIG. **13**, and a second peripheral circuit chip PC**2**_DB of the memory device **200***f* of FIG. **17** may include components included in the I/O chip IOC of FIG. **13**.
[0153] Referring to FIG. **18**, the memory device **200***g* of FIG. **18** may correspond to the memory device **200***b* of FIG. **14**. However, the memory device **200***g* of FIG. **18** may not include an I/O chip IOC unlike the memory device **200***b* of FIG. **14**, and a second peripheral circuit chip PC**2**_DB of the memory device **200***g* of FIG. **18** may include components included in the I/O chip IOC of FIG. **14**.
[0154] Referring to FIG. **19**, the memory device **200***h* of FIG. **19** may correspond to the memory device **200***c* of FIG. **15**. However, the memory device **200***h* of FIG. **19** may not include an I/O chip IOC unlike the memory device **200***c* of FIG. **15**, and a second peripheral circuit chip PC**2**_DB of the memory device **200***h* of FIG. **19** may include components included in the I/O chip IOC of FIG. **15**.
[0155] Referring to FIG. **20**, the memory device **200***i* of FIG. **20** may correspond to the memory device **200***d* of FIG. **16**. However, the memory device **200***i* of FIG. **20** may not include an I/O chip IOC unlike the memory device **200***d* of FIG. **16**, and a second peripheral circuit chip PC**2**_DB of the memory device **200***h* of FIG. **20** may include components included in the I/O chip IOC of FIG. **16**.
[0156] In FIGS. **17** to **20**, the second peripheral circuit chip PC**2**_DB is illustrated as including components included in the I/O chip IOC of FIGS. **13** to **16**, and a first peripheral circuit chip PC**1** of FIGS. **17** to **20** may include components included in the I/O chip IOC of FIGS. **13** to **16**.
[0157] Referring to FIGS. **17** to **20**, the second peripheral circuit chip PC**2**_DB may include a second page buffer circuit PBC**2**, a first I/O circuit **301**, a second I/O circuit **302**, a first pad circuit **303**, and a second pad circuit **304**. The second page buffer circuit PBC**2** may be formed on a front side unit FS of the second peripheral circuit chip PC**2**_DB. The first I/O circuit **301**, the second I/O circuit **302**, the first pad circuit **303**, and the second pad circuit **304** may be formed on a back side unit BS of the second peripheral circuit chip PC**2**_DB.
[0158] Referring to FIGS. **17** and **19**, the first peripheral circuit chip PC**1** may be electrically connected to the back side unit BS of the second peripheral circuit chip PC**2**_DB through a first connection unit CN**1***c* and a second connection unit CN**2***c*.
[0159] The first connection unit CN**1***c* may include a TSV TSV**11** and a TSV connection unit BP**11**. The TSV TSV**11** may pass through the first peripheral circuit chip PC**1** and be electrically connected to the TSV connection unit BP**11**. The TSV connection unit BP**11** may be electrically connected to the first I/O circuit **301**.
[0160] The second connection unit CN**2***c* may include a TSV TSV**12** and a TSV connection unit BP**12**. The TSV TSV**12** may pass through the first peripheral circuit chip PC**1** and be electrically connected to the TSV connection unit BP**12**. The TSV connection unit BP**12** may be electrically connected to the second I/O circuit **302**. A TSV TSV**22** may pass through the second peripheral circuit chip PC**2**_DB and be electrically connected to a TSV connection unit BP**22**.
[0161] The front side unit FS and the back side unit BS of the second peripheral circuit chip

PC**2**_DB may be electrically connected to each other through a third connection unit CN**3***c* and a fourth connection unit CN**4***c*.
[0162] The third connection unit CN**3***c* may include a TSV TSV**21** and a TSV connection unit BP**21**. The TSV TSV**21** may pass through the second peripheral circuit chip PC**2**_DB and be electrically connected to the TSV connection unit BP**21**. The TSV connection unit BP**21** may be electrically connected to the first I/O circuit **301**.
[0163] The fourth connection unit CN**4***c* may electrically connect the front side unit FS and the back side unit BS of the second peripheral circuit chip PC**2**_DB to each other. The fourth connection unit CN**4***c* may include the TSV TSV**22** and the TSV connection unit BP**22**. The TSV TSV**22** may pass through the second peripheral circuit chip PC**2**_DB and be electrically connected to the TSV connection unit BP**22**. The TSV connection unit BP**22** may be electrically connected to the second I/O circuit **302**.
[0164] Referring to FIGS. **18** and **20**, the first peripheral circuit chip PC**1** and the second peripheral circuit chip PC**2**_DB may be electrically connected to each other through a first connection unit CN**1***d* and a second connection unit CN**2***d*.
[0165] The first connection unit CN**1***d* may include a TSV TSV**11** and a TSV connection unit BP**11**. The TSV TSV**11** may be electrically connected to the TSV connection unit BP**11** through the first peripheral circuit chip PC**1**. The TSV connection unit BP**11** may be electrically connected to a first I/O circuit **301**.
[0166] The second connection unit CN**2***d* may include a TSV TSV**22** and a TSV connection unit BP**22**. The TSV TSV**22** may pass through the second peripheral circuit chip PC**2**_DB and be electrically connected to the TSV connection unit BP**22**. The TSV connection unit BP**22** may be electrically connected to a second I/O circuit **302**.
[0167] FIG. **21** is a view illustrating a memory device **200***j* according to embodiments.
[0168] Referring to FIG. **21**, the memory device **200***j* may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, a second cell array chip CAC**2**, a second peripheral circuit chip PC**2**, an I/O chip IOC, a third peripheral circuit chip PC**3**, a third cell array chip CAC**3**, a fourth peripheral circuit chip PC**4**, and a fourth cell array chip CAC**4**. Also, the memory device **200***j* may include a first bonding pad PD**1**, a second bonding pad PD**2**, metal patterns MP**11**, MP**12**, MP**21**, MP**22**, MP**31**, MP**32**, MP**41**, and MP**42**, contact plugs PT**11**, PT**12**, PT**21**, PT**22**, PT**31**, PT**32**, PT**41**, and PT**42**, contact bonding pads PBP**11**, PBP**12**, PBP**21**, PBP**22**, PBP**31**, PBP**32**, PBP**41**, and PBP**42**, TSVs TSV**11**, TSV**12**, TSV**21**, TSV**22**, TSV**31**, TSV**32**, TSV**33**, TSV**34**, TSV**41**, TSV**42**, TSV**43**, TSV**44**, TSV**51**, TSV**52**, TSV**61**, and TSV**62**, and TSV connection units BP**11**, BP**12**, BP**21**, BP**22**, BP**31**, BP**32**, BP**33**, BP**34**, BP**41**, BP**42**, BP**43**, BP**44**, BP**51**, BP**52**, BP**61**, and BP**62**.
[0169] The contact plug PT**11** and the contact plug PT**12** may pass through the first cell array chip CAC**1**. The contact plug PT**21** and the contact plug PT**22** may pass through the first peripheral circuit chip PC**1**. The contact plug PT**31** and the contact plug PT**32** may pass through the second cell array chip CAC**2**. The contact plug PT**41** and the contact plug PT**42** may pass through the second peripheral circuit chip PC**2**.
[0170] The first bonding pad PD**1** may be electrically connected to the contact bonding pad PBP**11** through the contact plug PT**11**. The second bonding pad PD**2** may be electrically connected to the contact bonding pad PBP**12** through the contact plug PT**12**. The contact plug PT**11** may be electrically connected to the contact plug PT**21** through the contact bonding pad PBP**11**. The contact plug PT**12** may be electrically connected to the contact plug PT**22** through the contact bonding pad PBP**12**. The contact plug PT**21** may be electrically connected to the contact plug PT**31** through the contact bonding pad PBP**21**. The contact plug PT**22** may be electrically connected to the contact plug PT**32** through the contact bonding pad PBP**22**. The contact plug PT**31** may be electrically connected to the contact plug PT**41** through the contact bonding pad PBP**31**. The contact plug PT**32** may be electrically connected to the contact plug PT**42** through the contact

bonding pad PBP**32**. The contact plug PT**41** may be electrically connected to the I/O chip IOC through the contact bonding pad PBP**41**. The contact plug PT**42** may be electrically connected to the I/O chip IOC through the contact bonding pad PBP**42**.

[0171] Components included in the first peripheral circuit chip PC**1** may transmit and receive data to and from the I/O chip IOC through the TSVs TSV**11**, TSV**12**, TSV**21**, TSV**22**, TSV**31**, and TSV**32**. The TSV TSV**11** and the TSV TSV**12** may pass through the first peripheral circuit chip PC**1**. The TSV TSV**21** and the TSV TSV**22** may pass through the first cell array chip CAC**1**. The TSV TSV**31** and the TSV TSV**32** may pass through the second peripheral circuit chip PC**2**.

[0172] The TSVs TSV**11** and TSV**12** may be electrically connected to components of the first peripheral circuit chip PC**1**. The TSV TSV**11** may be electrically connected to the TSV TSV**21** through the TSV connection unit BP**11**. The TSV TSV**12** may be electrically connected to the TSV TSV**22** through the TSV connection unit BP**12**. The TSV TSV**21** may be electrically connected to the TSV TSV**31** through the TSV connection unit BP**21**. The TSV TSV**22** may be electrically connected to the TSV TSV**32** through the TSV connection unit BP**22**. The TSV TSV**31** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**31**. The TSV TSV**32** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**32**.

[0173] Components included in the second peripheral circuit chip PC**2** may transmit and receive data to and from the I/O chip IOC through the TSVs TSV**33** and TSV**34**. The TSVs TSV**33** and TSV**34** may pass through the second peripheral circuit chip PC**2**. The TSV TSV**33** may electrically connect components included in the second peripheral circuit chip PC**2** to the TSV connection unit BP**33**. The TSV TSV**34** may electrically connect components included in the second peripheral circuit chip PC**2** to the TSV connection unit BP**34**. The TSV connection unit BP**33** may electrically connect the TSV TSV**33** to the I/O chip IOC. The TSV connection unit BP**34** may electrically connect the TSV TSV**34** to the I/O chip IOC.

[0174] Components include in the third peripheral circuit chip PC**3** may transmit and receive data to and from the I/O chip IOC through the TSVs TSV**43** and TSV**44**. The TSVs TSV**43** and TSV**44** may pass through the third peripheral circuit chip PC**3**. The TSV TSV**43** may electrically connect components included in the third peripheral circuit chip PC**3** to the TSV connection unit BP**43**. The TSV TSV**44** may electrically connect the components included in the third peripheral circuit chip PC**3** to the TSV connection unit BP**44**. The TSV connection unit BP**43** may electrically connect the TSV TSV**43** to the I/O chip IOC. The TSV connection unit BP**34** may electrically connect the TSV TSV**34** to the I/O chip IOC.

[0175] Components included in the fourth peripheral circuit chip PC**4** may transmit and receive data to and from the I/O chip IOC through the TSVs TSV**41**, TSV**42**, TSV**51**, TSV**52**, TSV**61**, and TSV**62**. The TSV TSV**41** and the TSV TSV**42** may pass through the third peripheral circuit chip PC**3**. The TSV TSV**51** and the TSV TSV**52** may pass through the third cell array chip CAC**3**. The TSV TSV**61** and the TSV TSV**62** may pass through the fourth peripheral circuit chip PC**4**.

[0176] The TSV TSV**41** may be electrically connected to the TSV TSV**51** through the TSV connection unit BP**51**. The TSV TSV**42** may be electrically connected to the TSV TSV**52** through the TSV connection unit BP**52**. The TSV TSV**51** may be electrically connected to the TSV TSV**61** through the TSV connection unit BP**61**. The TSV TSV**52** may be electrically connected to the TSV TSV**62** through the TSV connection unit BP**62**. The TSVs TSV**61** and TSV**62** may be electrically connected to components of the fourth peripheral circuit chip PC**4**.

[0177] The first bonding pad PD**1** and the second bonding pad PD**2** may be disposed on the first cell array chip CAC**1**. The first peripheral circuit chip PC**1** may be under the first cell array chip CAC**1**. The second cell array chip CAC**2** may be under the first peripheral circuit chip PC**1**. The second peripheral circuit chip PC**2** may be under the second cell array chip CAC**2**. The I/O chip IOC may be under the second peripheral circuit chip PC**2**. The third peripheral circuit chip PC**3** may be disposed under the I/O chip IOC. The third cell array chip CAC**3** may be under the third peripheral circuit chip PC**3**. The fourth peripheral circuit chip PC**4** may be under the third cell array

chip CAC**3**. The fourth cell array chip CAC**4** may be under the fourth peripheral circuit chip PC**4**.
[0178] The first cell array chip CAC**1** may be bonded to the first peripheral circuit chip PC**1** through the metal patterns MP**11** and MP**12**. The second cell array chip CAC**2** may be bonded to the second peripheral circuit chip PC**2** through the metal patterns MP**21** and MP**22**. The third cell array chip CAC**3** may be bonded to the third peripheral circuit chip PC**3** through the metal patterns MP**31** and MP**32**. The fourth cell array chip CAC**4** may be bonded to the fourth peripheral circuit chip PC**4** through the metal patterns MP**41** and MP**42**.
[0179] The first peripheral circuit chip PC**1** may be electrically connected to the I/O chip IOC through a first connection unit and a second connection unit. Each of the first connection unit and the second connection unit may include a TSV and pass through the first peripheral circuit chip PC**1**, the second cell array chip CAC**2**, and the second peripheral circuit chip PC**2**.
[0180] In an embodiment, the I/O chip IOC may further include a buffer chip. The buffer chip may be a chip configured to interface signals transmitted and received between the memory controller **100** and the first cell array chip CAC**1**, the second cell array chip CAC**2**, the third cell array chip CAC**3**, and the fourth cell array chip CAC**4** to reduce load on the I/O chip IOC.
[0181] FIG. **22** is a view illustrating a memory device **200***k* according to embodiments. FIG. **22** may be described with reference to FIG. **21**, and repeated descriptions may be omitted.
[0182] The description below focuses on differences between the memory device **200***k* of FIG. **22** and the memory device **200***j* of FIG. **21**.
[0183] Referring to FIG. **22**, the memory device **200***k* may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, a second cell array chip CAC**2**, a second peripheral circuit chip PC**2**, an I/O chip IOC, a third peripheral circuit chip PC**3**, a third cell array chip CAC**3**, a fourth peripheral circuit chip PC**4**, and a fourth cell array chip CAC**4**. Also, the memory device **200***k* may include a first bonding pad PD**1**, a second bonding pad PD**2**, metal patterns MP**11**, MP**12**, MP**21**, MP**22**, MP**31**, MP**32**, MP**41**, and MP**42**, contact plugs PT**11**, PT**12**, PT**21**, PT**22**, PT**31**, PT**32**, PT**41**, and PT**42**, contact bonding pads PBP**11**, PBP**12**, PBP**21**, PBP**22**, PBP**31**, PBP**32**, PBP**41**, and PBP**42**, TSVs TSV**11**, TSV**12**, TSV**21**, TSV**22**, TSV**31**, TSV**32**, TSV**41**, TSV**42**, TSV**51**, TSV**52**, TSV**61**, and TSV**62**, and TSV connection units BP**11**, BP**12**, BP**21**, BP**22**, BP**31**, BP**32**, BP**41**, BP**42**, BP**51**, BP**52**, BP**61**, and BP**62**.
[0184] The first peripheral circuit chip PC**1** may transmit and receive data to and from the I/O chip IOC through the second peripheral circuit chip PC**2**. Components included in the first peripheral circuit chip PC**1** may transmit and receive data to and from the second peripheral circuit chip PC**2** through the TSVs TSV**11**, TSV**12**, TSV**21**, and TSV**22**.
[0185] The TSVs TSV**11** and TSV**12** may pass through the first peripheral circuit chip PC**1**. The TSVs TSV**21** and TSV**22** may pass through the first cell array chip CAC**1**. The TSVs TSV**31**, TSV**32**, TSV**33**, and TSV**34** may pass through the second peripheral circuit chip PC**2**.
[0186] The TSVs TSV**11** and TSV**12** may be electrically connected to components of the first peripheral circuit chip PC**1**. The TSV TSV**11** may be electrically connected to the TSV TSV**21** through the TSV connection unit BP**11**. The TSV TSV**12** may be electrically connected to the TSV TSV**22** through the TSV connection unit BP**12**. The TSV TSV**21** may be electrically connected to the second peripheral circuit chip PC**2** through the TSV connection unit BP**21**. The TSV TSV**22** may be electrically connected to the second peripheral circuit chip PC**2** through the TSV connection unit BP**22**.
[0187] The TSV TSV**31** and the TSV TSV**32** may pass through the second peripheral circuit chip PC**2**. The TSVs TSV**31** and TSV**32** may be electrically connected to components of the second peripheral circuit chip PC**2**. The TSV TSV**31** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**31**. The TSV TSV**32** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**32**.
[0188] The TSV TSV**41** and the TSV TSV**42** may pass through the third peripheral circuit chip PC**3**. The TSVs TSV**41** and TSV**42** may be electrically connected to components of the third

peripheral circuit chip PC**3**. The TSV TSV**41** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**41**. The TSV TSV**42** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**42**.

[0189] The fourth peripheral circuit chip PC**4** may transmit and receive data to and from the I/O chip IOC through the third peripheral circuit chip PC**3**. Components included in the fourth peripheral circuit chip PC**4** may transmit and receive data to and from the third peripheral circuit chip PC**3** through the TSVs TSV**51**, TSV**52**, TSV**61**, and TSV**62**.

[0190] The TSVs TSV**41** and TSV**42** may pass through the third peripheral circuit chip PC**3**. The TSVs TSV**51** and TSV**52** may pass through the third cell array chip CAC**3**. The TSVs TSV**61** and TSV**62** may pass through the fourth peripheral circuit chip PC**4**.

[0191] The TSV TSV**51** may be electrically connected to the third peripheral circuit chip PC**3** through the TSV connection unit BP**51**. The TSV TSV**52** may be electrically connected to the third peripheral circuit chip PC**3** through the TSV connection unit BP**52**.

[0192] The TSV TSV**51** may be electrically connected to the TSV TSV**61** through the TSV connection unit BP**61**. The TSV TSV**52** may be electrically connected to the TSV TSV**62** through the TSV connection unit BP**62**. The TSVs TSV**61** and TSV**62** may be electrically connected to components of the fourth peripheral circuit chip PC**4**.

[0193] In an embodiment, the memory device **200***k* may not include the I/O chip IOC. In this case, the third peripheral circuit chip PC**3** of the memory device **200***k* may have the same structure as the second peripheral circuit chip PC**2**_DB shown in FIGS. **17** to **20**. That is, a page buffer circuit may be formed on a front surface of the third peripheral circuit chip PC**3**, and the first I/O circuit **301**, the second I/O circuit **302**, the first pad pad circuit **303**, and the second pad circuit **304** may be formed on a rear surface of the third peripheral circuit chip PC**3**.

[0194] FIG. **23** is a view illustrating a memory device **2001** according to embodiments. FIG. **23** may be described with reference to FIG. **21**, and repeated descriptions may be omitted.

[0195] The description below focuses on differences between the memory device **2001** of FIG. **23** and the memory device **200***j* of FIG. **21**.

[0196] Referring to FIG. **23**, the memory device **2001** may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, an I/O chip IOC, a second peripheral circuit chip PC**2**, a second cell array chip CAC**2**, a third peripheral circuit chip PC**3**, a third cell array chip CAC**3**, a first bonding pad PD**1**, a second bonding pad PD**2**, metal patterns MP**11**, MP**12**, MP**21**, MP**22**, MP**31**, and MP**32**, contact plugs PT**11**, PT**12**, PT**21**, and PT**22**, contact bonding pads PBP**11**, PBP**12**, PBP**21**, and PBP**22**, TSVs TSV**11**, TSV**12**, TSV**21**, TSV**22**, TSV**23**, TSV**24**, TSV**31**, TSV**32**, TSV**41**, and TSV**42**, and TSV connection units BP**11**, BP**12**, BP**21**, BP**22**, BP**23**, BP**24**, BP**31**, BP**32**, BP**41**, and BP**42**.

[0197] Chips included in the memory device **2001** may be asymmetrically arranged about the I/O chip IOC. Specifically, the first peripheral circuit chip PC**1** may be under the first cell array chip CAC**1**. The I/O chip IOC may be under the first peripheral circuit chip PC**1**. The second peripheral circuit chip PC**2** may be under the I/O chip IOC. The second cell array chip CAC**2** may be under the second peripheral circuit chip PC**2**. The third peripheral circuit chip PC**3** may be under the second cell array chip CAC**2**. The third cell array chip CAC**3** may be under the third peripheral circuit chip PC**3**.

[0198] The first cell array chip CAC**1** may be bonded to the first peripheral circuit chip PC**1** through the metal patterns MP**11** and MP**12**. The second cell array chip CAC**2** may be bonded to the second peripheral circuit chip PC**2** through the metal patterns MP**21** and MP**22**. The third cell array chip CAC**3** may be bonded to the third peripheral circuit chip PC**3** through the metal patterns MP**31** and MP**32**.

[0199] The contact plug PT**11** and the contact plug PT**12** may pass through the first cell array chip CAC**1**. The contact plug PT**21** and the contact plug PT**22** may pass through the first peripheral circuit chip PC**1**.

[0200] The first bonding pad PD**1** may be electrically connected to the contact bonding pad PBP**11** through the contact plug PT**11**. The second bonding pad PD**2** may be electrically connected to the contact bonding pad PBP**12** through the contact plug PT**12**. The contact plug PT**11** may be electrically connected to the contact plug PT**21** through the contact bonding pad PBP**11**. The contact plug PT**12** may be electrically connected to the contact plug PT**22** through the contact bonding pad PBP**12**. The contact plug PT**21** may be electrically connected to the I/O chip IOC through the contact bonding pad PBP**21**. The contact plug PT**22** may be electrically connected to the I/O chip IOC through the contact bonding pad PBP**22**.
[0201] The TSVs TSV**11** and TSV**12** may pass through the first peripheral circuit chip PC**1**. The TSVs TSV**21**, TSV**22**, TSV**23**, and TSV**24** may pass through the second peripheral circuit chip PC**2**. The TSVs TSV**31** and TSV**32** may pass through the second cell array chip CAC**2**. The TSVs TSV**41** and TSV**42** may pass through the third peripheral circuit chip PC**3**.
[0202] The TSVs TSV**11** and TSV**12** may be electrically connected to components included in the first peripheral circuit chip PC**1**. The TSV TSV**11** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**11**. The TSV TSV**12** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**12**.
[0203] The TSV TSV**21** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**21**. The TSV TSV**22** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**22**. The TSV TSV**23** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**23**. The TSV TSV**24** may be electrically connected to the I/O chip IOC through the TSV connection unit BP**24**.
[0204] The TSVs TSV**23** and TSV**24** may be electrically connected to components included in the second peripheral circuit chip PC**2**. The TSV TSV**21** may be electrically connected to the TSV TSV**31** through the TSV connection unit BP**31**. The TSV TSV**22** may be electrically connected to the TSV TSV**32** through the TSV connection unit BP**32**. The TSV TSV**31** may be electrically connected to the TSV TSV**41** through the TSV connection unit BP**41**. The TSV TSV**32** may be electrically connected to the TSV TSV**42** through the TSV connection unit BP**42**. The TSVs TSV**41** and TSV**42** may be electrically connected to components included in the third peripheral circuit chip PC**3**.
[0205] In an embodiment, the memory device **2001** may not include the I/O chip IOC. In this case, the second peripheral circuit chip PC**2** of the memory device **2001** may have the same structure as the second peripheral circuit chip PC**2**_DB shown in FIGS. **17** to **20**. That is, a page buffer circuit may be formed on a front surface of the second peripheral circuit chip PC**2**, and the first I/O circuit **301**, the second I/O circuit **302**, a first pad circuit **303**, and a second pad circuit **304** may be formed on a rear surface of the second peripheral circuit chip PC**2**.
[0206] FIG. **24** is a view illustrating a memory device **200***m* according to embodiments.
[0207] Referring to FIG. **24**, the memory device **200***m* may include a first cell array chip CAC**1**, a first peripheral circuit chip PC**1**, a second peripheral circuit chip PC**2**, a second cell array chip CAC**2**, a first bonding pad PD**1**, a second bonding pad PD**2**, metal patterns MP**11**, MP**12**, MP**21**, and MP**22**, contact plugs PT**11**, PT**12**, and PT**22**, and contact bonding pads PBP**11**, PBP**12**, and PBP**22**.
[0208] The first cell array chip CAC**1** may be bonded to the first peripheral circuit chip PC**1** through the metal patterns MP**11** and MP**12**. The second cell array chip CAC**2** may be bonded to the second peripheral circuit chip PC**2** through the metal patterns MP**21** and MP**22**.
[0209] The memory device **200***m* may be divided into a first memory device **200***m*_**1** and a second memory device **200***m*_**2**. The first memory device **200***m*_**1** may include the first cell array chip CAC**1** and the first peripheral circuit chip PC**1**. The second memory device **200***m*_**2** may include the second cell array chip CAC**2** and a second peripheral circuit chip PC**2**.
[0210] The first peripheral circuit chip PC**1** may include a page buffer circuit configured to write data to a memory cell array of the first cell array chip CAC**1** or read data from the memory cell

array of the first cell array chip CAC**1**. The first peripheral circuit chip PC**1** may include an I/O circuit configured to perform an I/O operation related to the first cell array chip CAC**1**.

[0211] The second peripheral circuit chip PC**2** may include a page buffer circuit configured to write data to a memory cell array of the second cell array chip CAC**2** or read data from the memory cell array of the second cell array chip CAC**2**. The second peripheral circuit chip PC**2** may include an I/O circuit configured to an I/O operation related to the second cell array chip CAC**2**.

[0212] The contact plugs PT**11** and PT**12** may pass through the first cell array chip CAC**1**. The contact plug PT**22** may pass through the first peripheral circuit chip PC**1**.

[0213] The contact plug PT**11** may electrically connect the first bonding pad PD**1** to the contact bonding pad PBP**11**. The contact plug PT**12** may electrically connect the second bonding pad PD**2** to the contact bonding pad PBP**12**.

[0214] The contact bonding pad PBP**11** may electrically connect the contact plug PT**11** to the first peripheral circuit chip PC**1**. The contact bonding pad PBP**12** may electrically connect the contact plug PT**11** to the contact plug PT**22**. The contact plug PT**22** may be electrically connected to the second peripheral circuit chip PC**2** through the contact bonding pad PBP**22**.

[0215] The first memory device **200***m*_**1** may transmit and receive data to and from a memory controller **100** through the first bonding pad PD**1**. The second memory device **200***m*_**2** may transmit and receive data to and from the memory controller **100** through the second bonding pad PD**2**.

[0216] In an embodiment, the first memory device **200***m*_**1** may write data to the memory cell array of the first cell array chip CAC**1** or read data from the memory cell array of the first cell array chip CAC**1** through the first bonding pad PD**1**.

[0217] In an embodiment, the second memory device **200***m*_**2** may write data to the memory cell array of the second cell array chip CAC**2** or read data from the memory cell array of the second ell array chip CAC**2** through the second bonding pad PD**2**.

[0218] FIG. **25** is a view illustrating a memory device **1500** according to some embodiments of the present disclosure.

[0219] Referring to FIG. **25**, the memory device **1500** may have a chip-to-chip (C2C) structure. At least one upper chip including a cell region and a lower chip including a peripheral circuit region PERI may be manufactured separately, and then, the at least one upper chip and the lower chip may be connected to each other by a bonding method to realize the C2C structure. For example, the bonding method may mean a method of electrically or physically connecting a bonding metal pattern formed in an uppermost metal layer of the upper chip to a bonding metal pattern formed in an uppermost metal layer of the lower chip. For example, in a case in which the bonding metal patterns are formed of copper (Cu), the bonding method may be a Cu—Cu bonding method. Alternatively, the bonding metal patterns may be formed of aluminum (Al) or tungsten (W).

[0220] The memory device **1500** may include the at least one upper chip including the cell region. For example, as illustrated in FIG. **25**, the memory device **1500** may include two upper chips. However, the number of the upper chips is not limited thereto. In the case in which the memory device **1500** includes the two upper chips, a first upper chip including a first cell region CELL**1**, a second upper chip including a second cell region CELL**2**, and the lower chip including the peripheral circuit region PERI may be manufactured separately, and then, the first upper chip, the second upper chip and the lower chip may be connected to each other by the bonding method to manufacture the memory device **1500**. The first upper chip may be turned over and then may be connected to the lower chip by the bonding method, and the second upper chip may also be turned over and then may be connected to the first upper chip by the bonding method. Hereinafter, upper and lower portions of each of the first and second upper chips will be defined based on before each of the first and second upper chips is turned over. In other words, an upper portion of the lower chip may mean an upper portion defined based on a +Z-axis direction, and the upper portion of each of the first and second upper chips may mean an upper portion defined based on a-Z-axis direction in FIG. **25**. However, embodiments of the present disclosure are not limited thereto. In certain embodiments, one of the first upper chip and the second upper chip may be turned over and then may be connected to a corresponding chip by the bonding method.

[0221] Each of the peripheral circuit region PERI and the first cell region CELL**1** and the second cell region CELL**2** of the memory device **1500** may include an external pad bonding region PA, a word line bonding region WLBA, and a bit line bonding region BLBA.

[0222] The peripheral circuit region PERI may include a first substrate **1210** and a plurality of circuit elements **1220***a*, **1220***b* and **1220***c* formed on the first substrate **1210**. An interlayer insulating layer **1215** including one or more insulating layers may be provided on the plurality of circuit elements **1220***a*, **1220***b* and **1220***c*, and a plurality of metal lines electrically connected to the plurality of circuit elements **1220***a*, **1220***b* and **1220***c* may be provided in the interlayer insulating layer **1215**. For example, the plurality of metal lines may include first metal lines **1230***a*, **1230***b* and **1230***c* connected to the plurality of circuit elements **1220***a*, **1220***b* and **1220***c*, and second metal lines **1240***a*, **1240***b* and **1240***c* formed on the first metal lines **1230***a*, **1230***b* and **1230***c*. The plurality of metal lines may be formed of at least one from among various conductive materials. For example, the first metal lines **1230***a*, **1230***b* and **1230***c* may be formed of tungsten having a relatively high electrical resistivity, and the second metal lines **1240***a*, **1240***b* and **1240***c* may be formed of copper having a relatively low electrical resistivity.

[0223] The first metal lines **1230***a*, **1230***b* and **1230***c* and the second metal lines **1240***a*, **1240***b* and **1240***c* are illustrated and described in the present embodiments. However, embodiments of the present disclosure are not limited thereto. In certain embodiments, at least one or more additional metal lines may further be formed on the second metal lines **1240***a*, **1240***b* and **1240***c*. In this case, the second metal lines **1240***a*, **1240***b* and **1240***c* may be formed of aluminum, and at least some of the additional metal lines formed on the second metal lines **1240***a*, **1240***b* and **1240***c* may be formed of copper having an electrical resistivity lower than an electrical resistivity of aluminum of the second metal lines **1240***a*, **1240***b* and **1240***c*.

[0224] The interlayer insulating layer **1215** may be disposed on the first substrate **1210** and may include an insulating material such as silicon oxide and/or silicon nitride.

[0225] Each of the first cell region CELL**1** and the second cell region CELL**2** may include at least one memory block. The first cell region CELL**1** may include a second substrate **1310** and a common source line **1320**. A plurality of word lines **1330** (e.g., word lines **1331** to **1338**) may be stacked on the second substrate **1310** in a direction (i.e., the Z-axis direction) perpendicular to a top surface of the second substrate **1310**. String selection lines and a ground selection line may be disposed on and under the word lines **1330**, and the plurality of word lines **1330** may be disposed between the string selection lines and the ground selection line. Likewise, the second cell region CELL**2** may include a third substrate **1410** and a common source line **1420**, and a plurality of word lines **1430** (e.g., word lines **1431** to **1438**) may be stacked on the third substrate **1410** in a direction (i.e., the Z-axis direction) perpendicular to a top surface of the third substrate **1410**. Each of the second substrate **1310** and the third substrate **1410** may be formed of at least one from among various materials and may be, for example, a silicon substrate, a silicon-germanium substrate, a germanium substrate, or a substrate having a single-crystalline epitaxial layer grown on a single-crystalline silicon substrate. A plurality of channel structures CH may be formed in each of the first cell region CELL**1** and the second cell region CELL**2**.

[0226] In some embodiments, as illustrated in a region Al corresponding to a region A, a channel structure CH may be provided in the bit line bonding region BLBA and may extend in the direction perpendicular to the top surface of the second substrate **1310** to penetrate the word lines **1330**, the string selection lines, and the ground selection line. The channel structure CH may include a data storage layer, a channel layer, and a filling insulation layer. The channel layer may be electrically connected to a first metal line **1350***c* and a second metal line **1360***c* in the bit line bonding region BLBA. For example, the second metal line **1360***c* may be a bit line and may be connected to the channel structure CH through the first metal line **1350***c*. The second metal line **1360***c* (e.g., a bit line) may extend in a first direction (e.g., a Y-axis direction) parallel to the top surface of the second substrate **1310**.

[0227] In some embodiments, as illustrated in a region A**2** corresponding to a region A, the channel structure CH may include a lower channel LCH and an upper channel UCH, which are connected to each other. For example, the channel structure CH may be formed by a process of forming the lower channel LCH and a process of forming the upper channel UCH. The lower channel LCH may extend in the direction perpendicular to the top surface of the second substrate **1310** to penetrate the common source line **1320** and word lines **1331** and **1332** (e.g., lower word lines). The lower channel LCH may include a data storage layer, a channel layer, and a filling insulation layer and may be connected to the upper channel UCH. The upper channel UCH may penetrate word lines **1333** to **1338** (e.g., upper word lines). The upper channel UCH may include a data storage layer, a channel layer, and a filling insulation layer, and the channel layer of the upper channel UCH may be electrically connected to the first metal line **1350***c* and the second metal line **1360***c*. As a length of a channel increases, due to characteristics of manufacturing processes, it may be difficult to form a channel having a substantially uniform width. The memory device **1500** according to the present embodiments may include a channel having improved width uniformity due to the lower channel LCH and the upper channel UCH which are formed by the processes performed sequentially.

[0228] In the case in which the channel structure CH includes the lower channel LCH and the upper channel UCH as illustrated in the region A**2**, a word line located near to a boundary between the lower channel LCH and the upper channel UCH may be a dummy word line. For example, the word lines **1332** and **1333** adjacent to the boundary between the lower channel LCH and the upper channel UCH may be the dummy word lines. In this case, data may not be stored in memory cells connected to the dummy word line. Alternatively, the number of pages corresponding to the memory cells connected to the dummy word line may be less than the number of pages corresponding to the memory cells connected to a general word line. A level of a voltage applied to the dummy word line may be different from a level of a voltage applied to the general word line, and thus it is possible to reduce an influence of a non-uniform channel width between the lower and upper channels LCH and UCH on an operation of the memory device.

[0229] Meanwhile, the number of the word lines **1331** and **1332** (e.g., lower word lines) penetrated by the lower channel LCH is less than the number of the word lines **1333** to **1338** (e.g., upper word lines) penetrated by the upper channel UCH in the region A**2**. However, embodiments of the present disclosure are not limited thereto. In certain embodiments, the number of the lower word lines penetrated by the lower channel LCH may be equal to or more than the number of the upper word lines penetrated by the upper channel UCH. In addition, structural features and connection relation of the channel structure CH disposed in the second cell region CELL**2** may be substantially the same as those of the channel structure CH disposed in the first cell region CELL**1**.

[0230] In the bit line bonding region BLBA, a first through-electrode THV**1** may be provided in the first cell region CELL**1**, and a second through-electrode THV**2** may be provided in the second cell region CELL**2**. As illustrated in FIG. **25**, the first through-electrode THV**1** may penetrate the common source line **1320** and the plurality of word lines **1330**. In certain embodiments, the first through-electrode THV**1** may further penetrate the second substrate **1310**. The first through-electrode THV**1** may include a conductive material. Alternatively, the first through-electrode THV**1** may include a conductive material surrounded by an insulating material. The second through-electrode THV**2** may have the same shape and structure as the first through-electrode THV**1**.

[0231] In some embodiments, the first through-electrode THV**1** and the second through-electrode THV**2** may be electrically connected to each other through a first through-metal pattern **1372***d* and a second through-metal pattern **1472***d*. The first through-metal pattern **1372***d* may be formed at a bottom end of the first upper chip including the first cell region CELL**1**, and the second through-metal pattern **1472***d* may be formed at a top end of the second upper chip including the second cell region CELL**2**. The first through-electrode THV**1** may be electrically connected to the first metal line **1350***c* and the second metal line **1360***c*. A lower via **1371***d* may be formed between the first through-electrode THV**1** and the first through-metal pattern **1372***d*, and an upper via **1471***d* may be formed between the second through-electrode THV**2** and the second through-metal pattern **1472***d*. The first through-metal pattern **1372***d* and the second through-metal pattern **1472***d* may be connected to each other by the bonding method.

[0232] In addition, in the bit line bonding region BLBA, an upper metal pattern **1252** may be formed in an uppermost metal layer of the peripheral circuit region PERI, and an upper metal pattern **1392** having the same shape as the upper metal pattern **1252** may be formed in an uppermost metal layer of the first cell region CELL**1**. The upper metal pattern **1392** of the first cell region CELL**1** and the upper metal pattern **1252** of the peripheral circuit region PERI may be electrically connected to each other by the bonding method. In the bit line bonding region BLBA, the second metal line **1360***c* (e.g., a bit line) may be electrically connected to a page buffer included in the peripheral circuit region PERI. For example, some of the circuit elements **1220***c* of the peripheral circuit region PERI may constitute the page buffer, and the second metal line **1360***c* (e.g., a bit line) may be electrically connected to the circuit elements **1220***c* constituting the page buffer through an upper bonding metal pattern **1370***c* of the first cell region CELL**1** and an upper bonding metal pattern **1270***c* of the peripheral circuit region PERI.

[0233] Referring continuously to FIG. **25**, in the word line bonding region WLBA, the word lines **1330** of the first cell region CELL**1** may extend in a second direction (e.g., an X-axis direction) parallel to the top surface of the second substrate **1310** and may be connected to a plurality of cell contact plugs **1340** (e.g., contact plugs **1341** to **1347**). First metal lines **1350***b* and second metal lines **1360***b* may be sequentially connected onto the cell contact plugs **1340** connected to the word lines **1330**. In the word line bonding region WLBA, the cell contact plugs **1340** may be connected to the peripheral circuit region PERI through upper bonding metal patterns **1370***b* of the first cell region CELL**1** and upper bonding metal patterns **1270***b* of the peripheral circuit region PERI.

[0234] The cell contact plugs **1340** may be electrically connected to a row decoder included in the peripheral circuit region PERI. For example, some of the circuit elements **1220***b* of the peripheral circuit region PERI may constitute the row decoder, and the cell contact plugs **1340** may be electrically connected to the circuit elements **1220***b* constituting the row decoder through the upper bonding metal patterns **1370***b* of the first cell region CELL**1** and the upper bonding metal patterns **1270***b* of the peripheral circuit region PERI. In some embodiments, an operating voltage of the circuit elements **1220***b* constituting the row decoder may be different from an operating voltage of the circuit elements **1220***c* constituting the page buffer. For example, the operating voltage of the circuit elements **1220***c* constituting the page buffer may be greater than the operating voltage of the circuit elements **1220***b* constituting the row decoder.

[0235] Likewise, in the word line bonding region WLBA, the word lines **1430** of the second cell region CELL**2** may extend in the second direction (e.g., the X-axis direction) parallel to the top surface of the third substrate **1410** and may be connected to a plurality of cell contact plugs **1440** (e.g., contact plugs **1441** to **1447**). The cell contact plugs **1440** may be connected to the peripheral circuit region PERI through an upper metal pattern of the second cell region CELL**2** and lower and upper metal patterns and a cell contact plug **1348** of the first cell region CELL**1**.

[0236] In the word line bonding region WLBA, the upper bonding metal patterns **1370***b* may be formed in the first cell region CELL**1**, and the upper bonding metal patterns **1270***b* may be formed in the peripheral circuit region PERI. The upper bonding metal patterns **1370***b* of the first cell region CELL**1** and the upper bonding metal patterns **1270***b* of the peripheral circuit region PERI may be electrically connected to each other by the bonding method. The upper bonding metal patterns **1370***b* and the upper bonding metal patterns **1270***b* may be formed of aluminum, copper, or tungsten.

[0237] In the external pad bonding region PA, a lower metal pattern **1371***e* may be formed in a lower portion of the first cell region CELL**1**, and an upper metal pattern **1472***a* may be formed in an upper portion of the second cell region CELL**2**. The lower metal pattern **1371***e* of the first cell region CELL**1** and the upper metal pattern **1472***a* of the second cell region CELL**2** may be connected to each other by the bonding method in the external pad bonding region PA. Likewise, an upper metal pattern **1372***a* may be formed in an upper portion of the first cell region CELL**1**, and an upper metal pattern **1272***a* may be formed in an upper portion of the peripheral circuit region PERI. The upper metal pattern **1372***a* of the first cell region CELL**1** and the upper metal pattern **1272***a* of the peripheral circuit region PERI may be connected to each other by the bonding method.
[0238] Common source line contact plugs **1380** and **1480** may be disposed in the external pad bonding region PA. The common source line contact plugs **1380** and **1480** may be formed of a conductive material such as a metal, a metal compound, and/or doped polysilicon. The common source line contact plug **1380** of the first cell region CELL**1** may be electrically connected to the common source line **1320**, and the common source line contact plug **1480** of the second cell region CELL**2** may be electrically connected to the common source line **1420**. A first metal line **1350***a* and a second metal line **1360***a* may be sequentially stacked on the common source line contact plug **1380** of the first cell region CELL**1**, and a first metal line **1450***a* and a second metal line **1460***a* may be sequentially stacked on the common source line contact plug **1480** of the second cell region CELL**2**.
[0239] Input/output pads (e.g., a first input/output pad **1205**, a second input/output pad **1405**, and a third input/output pad **1406**) may be disposed in the external pad bonding region PA. Referring to FIG. **25**, a lower insulating layer **1201** may cover a bottom surface of the first substrate **1210**, and a first input/output pad **1205** may be formed on the lower insulating layer **1201**. The first input/output pad **1205** may be connected to at least one of a plurality of the circuit elements **1220***a* disposed in the peripheral circuit region PERI through a first input/output contact plug **1203** and may be separated from the first substrate **1210** by the lower insulating layer **1201**. In addition, a side insulating layer may be disposed between the first input/output contact plug **1203** and the first substrate **1210** to electrically isolate the first input/output contact plug **1203** from the first substrate **1210**.
[0240] An upper insulating layer **1401** covering a top surface of the third substrate **1410** may be formed on the third substrate **1410**. A second input/output pad **1405** and/or a third input/output pad **1406** may be disposed on the upper insulating layer **1401**. The second input/output pad **1405** may be connected to at least one of the plurality of circuit elements **1220***a* disposed in the peripheral circuit region PERI through second input/output contact plugs **1403** and **1303**, and the third input/output pad **1406** may be connected to at least one of the plurality of circuit elements **1220***a* disposed in the peripheral circuit region PERI through third input/output contact plugs **1404** and **1304**.
[0241] In some embodiments, the third substrate **1410** may not be disposed in a region in which the input/output contact plug is disposed. For example, as illustrated in a region B, the third input/output contact plug **1404** may be separated from the third substrate **1410** in a direction parallel to the top surface of the third substrate **1410** and may penetrate an interlayer insulating layer **1415** of the second cell region CELL**2** so as to be connected to the third input/output pad **1406**. In this case, the third input/output contact plug **1404** may be formed by at least one of various processes.
[0242] In some embodiments, as illustrated in a region B**1** corresponding to the region B, the third input/output contact plug **1404** may extend in a third direction (e.g., the Z-axis direction), and a diameter of the third input/output contact plug **1404** may become progressively greater toward the upper insulating layer **1401**. In other words, a diameter of the channel structure CH described in the region Al may become progressively less toward the upper insulating layer **1401**, but the diameter

of the third input/output contact plug **1404** may become progressively greater toward the upper insulating layer **1401**. For example, the third input/output contact plug **1404** may be formed after the second cell region CELL**2** and the first cell region CELL**1** are bonded to each other by the bonding method.

[0243] In certain embodiments, as illustrated in a region B**2** corresponding to the region B, the third input/output contact plug **1404** may extend in the third direction (e.g., the Z-axis direction), and a diameter of the third input/output contact plug **1404** may become progressively less toward the upper insulating layer **1401**. In other words, like the channel structure CH, the diameter of the third input/output contact plug **1404** may become progressively less toward the upper insulating layer **1401**. For example, the third input/output contact plug **1404** may be formed together with the cell contact plugs **1440** before the second cell region CELL**2** and the first cell region CELL**1** are bonded to each other.

[0244] In certain embodiments, the input/output contact plug may overlap with the third substrate **1410**. For example, as illustrated in a region C, the second input/output contact plug **1403** may penetrate the interlayer insulating layer **1415** of the second cell region CELL**2** in the third direction (e.g., the Z-axis direction) and may be electrically connected to the second input/output pad **1405** through the third substrate **1410**. In this case, a connection structure of the second input/output contact plug **1403** and the second input/output pad **1405** may be realized by various methods.

[0245] In some embodiments, as illustrated in a region C**1** corresponding to the region C, an opening **1408** may be formed to penetrate the third substrate **1410**, and the second input/output contact plug **1403** may be connected directly to the second input/output pad **1405** through the opening **1408** formed in the third substrate **1410**. In this case, as illustrated in the region C**1**, a diameter of the second input/output contact plug **1403** may become progressively greater toward the second input/output pad **1405**. However, embodiments of the present disclosure are not limited thereto, and in certain embodiments, the diameter of the second input/output contact plug **1403** may become progressively less toward the second input/output pad **1405**.

[0246] In certain embodiments, as illustrated in a region C**2** corresponding to the region C, the opening **1408** penetrating the third substrate **1410** may be formed, and a contact **1407** may be formed in the opening **1408**. An end of the contact **1407** may be connected to the second input/output pad **1405**, and another end of the contact **1407** may be connected to the second input/output contact plug **1403**. Thus, the second input/output contact plug **1403** may be electrically connected to the second input/output pad **1405** through the contact **1407** in the opening **1408**. In this case, as illustrated in the region C**2**, a diameter of the contact **1407** may become progressively greater toward the second input/output pad **1405**, and a diameter of the second input/output contact plug **1403** may become progressively less toward the second input/output pad **1405**. For example, the second input/output contact plug **1403** may be formed together with the cell contact plugs **1440** before the second cell region CELL**2** and the first cell region CELL**1** are bonded to each other, and the contact **1407** may be formed after the second cell region CELL**2** and the first cell region CELL**1** are bonded to each other.

[0247] In certain embodiments illustrated in a region C**3** corresponding to the region C, a stopper **1409** may further be formed on a bottom end of the opening **1408** of the third substrate **1410**, as compared with the embodiments of the region C**2**. The stopper **1409** may be a metal line formed in the same layer as the common source line **1420**. Alternatively, the stopper **1409** may be a metal line formed in the same layer as at least one of the word lines **1430**. The second input/output contact plug **1403** may be electrically connected to the second input/output pad **1405** through the contact **1407** and the stopper **1409**.

[0248] Like the second input/output contact plug **1403** and the third input/output contact plug **1404** of the second cell region CELL**2**, a diameter of each of the second input/output contact plug **1303** and the third input/output contact plug **1304** of the first cell region CELL**1** may become progressively less toward the lower metal pattern **1371***e* or may become progressively greater toward the lower metal pattern **1371***e*.

[0249] Meanwhile, in some embodiments, a slit **1411** may be formed in the third substrate **1410**. For example, the slit **1411** may be formed at a certain position of the external pad bonding region PA. For example, as illustrated in a region D, the slit **1411** may be located between the second input/output pad **1405** and the cell contact plugs **1440** when viewed in a plan view. Alternatively, the second input/output pad **1405** may be located between the slit **1411** and the cell contact plugs **1440** when viewed in a plan view.

[0250] In some embodiments, as illustrated in a region D**1** corresponding to the region D, the slit **1411** may be formed to penetrate the third substrate **1410**. For example, the slit **1411** may be used to prevent the third substrate **1410** from being finely cracked when the opening **1408** is formed. However, embodiments of the present disclosure are not limited thereto, and in certain embodiments, the slit **1411** may be formed to have a depth ranging from about 60% to about 70% of a thickness of the third substrate **1410**.

[0251] In certain embodiments, as illustrated in a region D**2** corresponding to the region D, a conductive material **1412** may be formed in the slit **1411**. For example, the conductive material **1412** may be used to discharge a leakage current occurring in driving of the circuit elements in the external pad bonding region PA to the outside. In this case, the conductive material **1412** may be connected to an external ground line.

[0252] In certain embodiments, as illustrated in a region D**3** corresponding to the region D, an insulating material **1413** may be formed in the slit **1411**. For example, the insulating material **1413** may be used to electrically isolate the second input/output pad **1405** and the second input/output contact plug **1403** disposed in the external pad bonding region PA from the word line bonding region WLBA. Since the insulating material **1413** is formed in the slit **1411**, it is possible to prevent a voltage provided through the second input/output pad **1405** from affecting a metal layer disposed on the third substrate **1410** in the word line bonding region WLBA.

[0253] Meanwhile, in certain embodiments, the first input/output pad **1205**, the second input/output pad **1405**, and the third input/output pad **1406** may be selectively formed. For example, the memory device **1500** may be realized to include only the first input/output pad **1205** disposed on the first substrate **1210**, to include only the second input/output pad **1405** disposed on the third substrate **1410**, or to include only the third input/output pad **1406** disposed on the upper insulating layer **1401**.

[0254] In some embodiments, at least one from among the second substrate **1310** of the first cell region CELL**1** and the third substrate **1410** of the second cell region CELL**2** may be used as a sacrificial substrate and may be completely or partially removed before or after a bonding process. An additional layer may be stacked after the removal of the substrate. For example, the second substrate **1310** of the first cell region CELL**1** may be removed before or after the bonding process of the peripheral circuit region PERI and the first cell region CELL**1**, and then, an insulating layer covering a top surface of the common source line **1320** or a conductive layer for connection may be formed. Likewise, the third substrate **1410** of the second cell region CELL**2** may be removed before or after the bonding process of the first cell region CELL**1** and the second cell region CELL**2**, and then, the upper insulating layer **1401** covering a top surface of the common source line **1420** or a conductive layer for connection may be formed.

[0255] FIG. **26** is a block diagram of an example of applying a memory device according to embodiments to an SSD system **2000**.

[0256] Referring to FIG. **26**, the SSD system **2000** may include a host **2100** and an SSD **2200**. The SSD **2200** may transmit and receive a signal to and from the host **2100** through a signal connector and receive power through a power connector. The SSD **2200** may include an SSD controller **2210**, an auxiliary power supply **2220**, a first memory device **2230**, a second memory device **2240**, and a third memory device **2250**. Although the SSD **2200** is illustrated as including three memory devices as an example in FIG. **26**, the SSD **2200** may include fewer or more memory devices than in FIG. **26**.

[0257] Each of the first memory device **2230**, the second memory device **2240**, and the third memory device **2250** may be a vertical-stack-type NAND flash memory device. Each of the first memory device **2230**, the second memory device **2240**, and the third memory device **2250** may transmit and receive signals to the SSD controller **2210** through a first channel Ch**1**, a second channel Ch**2**, and a third channel Ch**3**. The first memory device **2230**, the second memory device **2240**, and the third memory device **2250** may be realized based on the embodiments described above with reference to FIGS. **1** to **25**.

[0258] While non-limiting example embodiments of the present disclosure have been particularly shown and described with reference to the drawings, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure.

Claims

1. A memory device comprising: a first cell array chip comprising a first memory cell array; a first peripheral circuit chip comprising a first page buffer circuit electrically connected to the first memory cell array; an input/output chip comprising a first pad circuit, a second pad circuit, a first input/output circuit electrically connected to the first pad circuit, and a second input/output circuit electrically connected to the second pad circuit; a second cell array chip comprising a second memory cell array; and a second peripheral circuit chip comprising a second page buffer circuit electrically connected to the second memory cell array.

2. The memory device of claim 1, further comprising: a first bonding pad configured to electrically connect an external device to the memory device; and a second bonding pad configured to electrically connect the external device to the memory device, wherein the first peripheral circuit chip is under the first cell array chip, wherein the input/output chip is under the first peripheral circuit chip, wherein the second peripheral circuit chip is under the input/output chip, wherein the second cell array chip is under the second peripheral circuit chip, and wherein the first bonding pad and the second bonding pad are on the first cell array chip.

3. The memory device of claim 2, wherein the memory device further comprises: first contact plugs passing through the first peripheral circuit chip and the first cell array chip, the first contact plugs configured to electrically connect the first pad circuit to the first bonding pad; and second contact plugs passing through the first peripheral circuit chip and the first cell array chip, the second contact plugs configured to electrically connect the second pad circuit to the second bonding pad.

4. The memory device of claim 1, further comprising: a first edge pad configured to electrically connect an external device to the memory device; and a second edge pad configured to electrically connect the external device to the memory device, wherein the first peripheral circuit chip is under the first cell array chip, wherein the input/output chip is under the first peripheral circuit chip, wherein the second peripheral circuit chip is under the input/output chip, wherein the second cell array chip is under the second peripheral circuit chip, and wherein the first edge pad and the second edge pad are between the first peripheral circuit chip and the second peripheral circuit chip.

5. The memory device of claim 2, wherein the memory device is configured to transmit and receive first data to and from the external device through the first bonding pad and the second bonding pad in response to a first command received from the external device.

6. The memory device of claim 2, wherein the memory device is configured to transmit and receive first data to and from the external device through the first bonding pad in response to a first command received from the external device, and transmit and receive second data to and from the external device through the second bonding pad in response to a second command received from the external device.

7. The memory device of claim 1, wherein the first memory cell array comprises a first memory region and a second memory region, wherein the second memory cell array comprises a third memory region and a fourth memory region, wherein the first page buffer circuit comprises a first page buffer overlapping with the first memory region in a vertical direction and a second page buffer overlapping with the second memory region in the vertical direction, and wherein the second page buffer circuit comprises a third page buffer overlapping with the third memory region in the vertical direction and a fourth page buffer overlapping with the fourth memory region in the vertical direction.

8. The memory device of claim 7, wherein the memory device is configured to: provide data stored in the first memory region to an external device through the first page buffer and the first input/output circuit; provide data stored in the second memory region to the external device through the second page buffer and the first input/output circuit; provide data stored in the third memory region to the external device through the third page buffer and the second input/output circuit; and provide data stored in the fourth memory region to the external device through the fourth page buffer and the second input/output circuit.

9. The memory device of claim 7, wherein the memory device is configured to: provide data stored in the first memory region to an external device through the first page buffer and the first input/output circuit; provide data stored in the third memory region to the external device through the third page buffer and the first input/output circuit; provide data stored in the second memory region to the external device through the second page buffer and the second input/output circuit; and provide data stored in the fourth memory region to the external device through the fourth page buffer and the second input/output circuit.

10. The memory device of claim 9, wherein a distance between the first memory region and the first input/output circuit is less than a distance between the second memory region and the first input/output circuit, and wherein a distance between the third memory region and the first input/output circuit is less than a distance between the fourth memory region and the first input/output circuit.

11. A memory device comprising: a first cell array chip comprising a first memory cell array; a first peripheral circuit chip comprising a first page buffer circuit electrically connected to the first memory cell array; a second cell array chip comprising a second memory cell array; and a second peripheral circuit chip comprising a second page buffer circuit, a first pad circuit, a second pad circuit, a first input/output circuit electrically connected to the first pad circuit, and a second input/output circuit electrically connected to the second pad circuit, wherein the second page buffer circuit is electrically connected to the second memory cell array and formed on one surface of the second peripheral circuit chip, and wherein the first pad circuit, the second pad circuit, the first input/output circuit, and the second input/output circuit are formed on another surface of the second peripheral circuit chip.

12. The memory device of claim 11, further comprising: a first bonding pad configured to electrically connect an external device to the memory device; and a second bonding pad configured to electrically connect the external device to the memory device, wherein the first peripheral circuit chip is under the first cell array chip, wherein the second peripheral circuit chip is under the first peripheral circuit chip, wherein the second cell array chip is under the second peripheral circuit chip, and wherein the first bonding pad and the second bonding pad are on the first cell array chip.

13. The memory device of claim 12, wherein the memory device further comprises: a first contact passing through the first peripheral circuit chip and the first cell array chip, the first contact configured to electrically connect the first pad circuit to the first bonding pad; and a second contact passing through the first peripheral circuit chip and the first cell array chip, the second contact configured to electrically connect the second pad circuit to the second bonding pad.

14. The memory device of claim 11, further comprising: a first edge pad configured to electrically connect an external device to the memory device; and a second edge pad configured to electrically connect the external device to the memory device, wherein the first peripheral circuit chip is under the first cell array chip, wherein the second peripheral circuit chip is under the first peripheral circuit chip, wherein the second cell array chip is under the second peripheral circuit chip, and wherein the first edge pad and the second edge pad are between the first peripheral circuit chip and the second peripheral circuit chip.

15. The memory device of claim 11, wherein the first memory cell array comprises a first memory region and a second memory region, wherein the second memory cell array comprises a third memory region and a fourth memory region, wherein the first page buffer circuit comprises a first page buffer overlapping with the first memory region in a vertical direction and a second page buffer overlapping with the second memory region in the vertical direction, and wherein the second page buffer circuit comprises a third page buffer overlapping with the third memory region in the vertical direction and a fourth page buffer overlapping with the fourth memory region in the vertical direction.

16. The memory device of claim 15, wherein the memory device is configured to: provide data stored in the first memory region to an external device through the first page buffer and the first input/output circuit; provide data stored in the third memory region to the external device through the third page buffer and the first input/output circuit; provide data stored in the second memory region to the external device through the second page buffer and the second input/output circuit; and provide data stored in the fourth memory region to the external device through the fourth page buffer and the second input/output circuit.

17. The memory device of claim 16, wherein a distance between the first memory region and the first input/output circuit is less than a distance between the second memory region and the first input/output circuit, and wherein a distance between the third memory region and the first input/output circuit is less than a distance between the fourth memory region and the first input/output circuit.

18. A memory device comprising: a first cell array chip comprising a first memory cell array; a first peripheral circuit chip comprising a first page buffer circuit electrically connected to the first memory cell array; an input/output chip comprising a first pad circuit, a second pad circuit, a first input/output circuit electrically connected to the first pad circuit, and a second input/output circuit electrically connected to the second pad circuit; a second cell array chip comprising a second memory cell array; a second peripheral circuit chip comprising a second page buffer circuit electrically connected to the second memory cell array; a third cell array chip comprising a third memory cell array; and a third peripheral circuit chip comprising a third page buffer circuit electrically connected to the third memory cell array.

19. The memory device of claim 18, further comprising: a first bonding pad configured to electrically connect an external device to the memory device; and a second bonding pad configured to electrically connect the external device to the memory device, wherein the first peripheral circuit chip is under the first cell array chip, wherein the input/output chip is under the first peripheral circuit chip, wherein the second peripheral circuit chip is under the input/output chip, wherein the second cell array chip is under the second peripheral circuit chip, wherein the third peripheral circuit chip is under the second cell array chip, wherein the third cell array chip is under the third peripheral circuit chip, wherein the first bonding pad and the second bonding pad are on the first cell array chip, and wherein the third peripheral circuit chip is electrically connected to the input/output chip through at least one through-silicon via (TSV).

20. The memory device of claim 19, further comprising: a first contact passing through the first peripheral circuit chip and the first cell array chip, the first contact configured to electrically connect the first pad circuit to the first bonding pad; and a second contact passing through the first peripheral circuit chip and the first cell array chip, the second contact configured to electrically connect the second pad circuit to the second bonding pad.